



MG2009 – DATA ANALYSIS FOR BUSINESS – II

DETERMINANTS OF SMARTPHONE PRICING

A Statistical Analysis of Hardware Specifications & Brand Equity

Final Project Report

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Abstract

This study aims to deconstruct the pricing mechanisms within the global smartphone industry by analysing the relationship between tangible hardware specifications and retail pricing. Using a quantitative research design, secondary data was collected from the Kaggle repository for 558 unique mobile devices across five major brands: Samsung, Apple, Xiaomi, OnePlus, and Google Pixel. The research employed descriptive statistics to profile market segments and inferential statistical methods, including One-Way ANOVA, Independent Samples T-Tests, and Multiple Linear Regression, using IBM SPSS Statistics to test five hypotheses.

Contrary to general market assumptions, the results indicated no statistically significant difference in mean prices across the selected brands. Furthermore, correlation and regression analyses revealed that hardware specifications such as RAM, battery capacity, and camera quality are not significant predictors of retail price in this dataset. The study also found that 5G connectivity does not command a statistically significant price premium compared to non-5G devices. However, consumer satisfaction was found to be significantly higher than the market benchmark. *The study concludes that smartphone pricing is likely driven by intangible factors such as brand equity and ecosystem integration rather than raw performance metrics.*

Acknowledgement

All praises and thanks are due to the **Almighty**, the source of all knowledge and wisdom, for granting us the strength and resilience to complete this research endeavour successfully.

We would like to express our profound gratitude to our distinguished lecturer, **Miss Aiza Aftab**, for her expert mentorship during the **Data Analysis for Business – II** course. Her ability to bridge the gap between theoretical concepts and real-world application provided the intellectual foundation necessary for this study on the "Determinants of Smartphone Pricing". Her critical feedback challenged us to look beyond the surface of the data and derive meaningful business insights.

We also owe a debt of gratitude to our Lab Instructor, **Miss Sadia Manzoor Malik**, whose technical guidance was instrumental to this project. Her hands-on approach to teaching **IBM SPSS Statistics** transformed a complex tool into an accessible asset for us. It was her patient instruction that empowered us to execute the advanced diagnostic and inferential statistical tests that form the core of this report.

Finally, we acknowledge **FAST School of Management** for providing a challenging academic environment that fosters research, critical thinking, and professional growth. This project has served not only as an academic requirement but as a significant milestone in our journey toward becoming data-driven business professionals.

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List of Abbreviations

5G	Fifth Generation (Mobile Network)
ANOVA	Analysis of Variance
CLT	Central Limit Theorem
GB	Giga Bytes
H ₀	Null Hypothesis
H ₁	Alternate Hypothesis
LTE	Long Term evolution
mAh	Milliampere- hour
mp	Mega Pixels
OS	Operating System
RAM	Random Access Memory
ROM	Read Only Memory
SPSS	Statistical Package for Social Sciences
USD	United States Dollars
WEF	World Economic Forum

Introduction & Literature Review

Stage 01 in the research

FINAL PROJECT REPORT

DETERMINANTS OF SMARTPHONE PRICINGS

A Statistical Analysis of Hardware Specifications & Brand Equity

1- Introduction & Literature Review:

Topic of the Research:

This research focuses on the **Determinants of Mobile Phone Pricing Strategies**, analyzing how technical specifications and brand equity influence market pricing. In the rapidly evolving telecommunications industry, mobile devices have transitioned from luxury items to essential tools, leading to a saturated market characterized by diverse pricing structures.

Understanding the relationship between hardware specifications such as RAM, battery capacity, and camera quality and the final retail price is crucial for understanding consumer value and market positioning.

Main Issues and problem need to be discussed:

The primary issue addressed in this study is the opacity of pricing mechanisms in the smartphone market. Consumers often struggle to discern whether a higher price tag reflects superior hardware performance or merely brand premium (Brand Equity).

Furthermore, with the introduction of new technologies such as 5G connectivity, there is a need to quantify the "premium" attached to these features. This study aims to deconstruct the price tag of modern smartphones to determine which factors: Brand, Memory, Battery, or Connectivity are the most significant drivers of cost.

Research Question:

To address these issues, this study will address or analyze the following questions:

1. Is there a statistically significant difference in pricing across different mobile phone brands?
2. What is the strength and direction of the relationship between hardware specifications (RAM, Battery, Camera) and Price?
3. Does the inclusion of 5G support significantly impact the average price of a mobile device?
4. Do consumer ratings reflect a standardized level of satisfaction across the market?
5. To what extent can the price of a mobile phone be predicted based on its internal specifications?

Source of the information:

The background information for this study is derived from **current market trends in the telecommunications sector**, posted by World Economic Forum Reports (WEF) in 2024.

The Secondary data for analysis was obtained from **Kaggle**, which aggregates technical specifications and market prices of current smartphone models.

The Dataset and the variables measured:

The scope of this research is specifically delimited to the analysis of five prominent mobile phone manufacturers, representing a significant share of the global market. The study focuses exclusively on the **following five brands**:

Table 1: Brands for Research Study

01	Samsung
02	Apple
03	Xiaomi
04	OnePlus
05	Google Pixels

The secondary dataset was acquired from the **Kaggle** data repository. The selected dataset has been pre-processed to ensuring it is cleaned, organized, and validated, resulting in a final sample size of **N=558** records.

The dataset comprises the following variables, categorized by their measurement scales and data types:

Table 2: Data Set description for Research Study

Variables	Datatypes/Measurement Scales
Brand	Categorical Strings
Model	Nominal Strings
Price (in usd)	Numerical
RAM (Gb)	Numerical
ROM (Gb)	Numerical
Camera (mp)	Numerical
Battery (mAh)	Numerical
Display size (inch)	Numerical
Charging (watt)	Numerical
5G Support	Boolean (Yes/No)
Operating System	Ordinal Strings (Android/ios)
Processor	Ordinal Strings
Rating	Ratio Scale

Objectives of the Research

Stage 01 (1.1) in the research

FINAL PROJECT REPORT

DETERMINANTS OF SMARTPHONE PRICINGS

A Statistical Analysis of Hardware Specifications & Brand Equity

1.1 – Objectives of the research:

Main Objective:

The primary objective of this research is to deconstruct the pricing mechanisms within the smartphone industry by identifying the key determinants of market value. The study aims to statistically analyze how tangible hardware specifications and intangible brand equity influence the final retail price, culminating in the development of a predictive pricing model.

Research Objectives:

To achieve the primary goal, this study focuses on the following five specific research objectives:

1. To evaluate whether statistically significant price disparities exist among the five major mobile phone brands.
2. To determine the strength and direction of the linear relationship between key technical specifications and the device Price.
3. To quantify the "5G Premium" by analysing if 5G-enabled devices command a significantly higher average price compared to non-5G devices.
4. To assess consumer satisfaction levels by comparing the average user rating of the sample against a market benchmark of 4.0.
5. To formulate a multiple linear regression model that predicts the price of a mobile phone based on its internal hardware components.
6. To identify feature patterns that distinguish high-priced, mid-range, and low-priced models based on descriptive statistical analysis.

Research Hypothesis:

In alignment with the specific objectives, the following null (H_0) and alternative (H_1) hypotheses have been formulated for statistical testing:

(Note: Objective 06 is exploratory and has been addressed through descriptive stats)

Hypothesis 01: (Brand Price Differentiation)

- **Null Hypothesis (H_0):** There is no significant difference in the mean price across different mobile phone brands.
- **Alternative Hypothesis (H_1):** There is a statistically significant difference in the mean price across different mobile phone brands.

Hypothesis 02: (Correlation of Specification with prices)

- **Null Hypothesis (H_0):** There is no significant relationship between hardware specifications and price.
- **Alternative Hypothesis (H_1):** There is a significant linear relationship between hardware specifications and price.

Hypothesis 03: (Impact of 5G connectivity on the prices)

- **Null Hypothesis (H_0):** There is no significant difference in the mean price between 5G Supported and non 5G devices.
- **Alternative Hypothesis (H_1):** The mean price of 5G supported phones is statistically significantly different from that of non-5g phones.

Hypothesis 04: (Consumer Rating Analysis against market benchmark score)

- **Null Hypothesis (H_0):** The average consumer rating for mobile phones is statistically equal to the market benchmark of 4.0.
- **Alternative Hypothesis (H_1):** The average consumer rating for mobile phones is significantly different from the market benchmark of 4.0.

Hypothesis 05: (Predictive Pricing Modelling)

- **Null Hypothesis (H_0):** The technical specifications do not significantly predict the prices of the mobile phones.
 - **Alternative Hypothesis (H_1):** The technical specifications can significantly predict the prices of the mobile phones.
-

Data Collection & Methodology

Stage 02 in the research

FINAL PROJECT REPORT

DETERMINANTS OF SMARTPHONE PRICINGS

A Statistical Analysis of Hardware Specifications & Brand Equity

2 – Data Collection and Methodology:

Research Design:

This study adopts a **Quantitative Research Design**, utilizing a positivist approach to objectively analyze market data. The research framework is primarily **Descriptive** and **Correlational-Causal** in nature. Descriptive methods are employed to profile the dataset and identify segmentation patterns (Objective 6), while correlational and causal techniques are utilized to test hypotheses regarding the relationships between technical specifications and pricing (Objectives 1–5).

Data Collection Strategy:

The study utilizes **Secondary Data** sourced from the **Kaggle** data repository. The dataset aggregates technical specifications and retail prices for current mobile phone models available in the global market:

- **Target Population:** Mobile phones manufactured by major global technology firms.
- **Sample Selection:** The study focuses on five specific brands: *Samsung, Apple, Xiaomi, OnePlus, and Google Pixel*.
- **Sample Size:** The final dataset consists of **N = 558** valid records after data cleaning and pre-processing.
- **Data Integrity:** The dataset was inspected for missing values and outliers. Categorical variables (*Brand, OS*) were coded appropriately for statistical analysis.

Statistical Analysis Tool:

All statistical analyses were conducted using **IBM SPSS Statistics (Statistical Package for the Social Sciences)**. The analysis was structured into three distinct phases:

- **Descriptive Analysis:** Computation of frequencies, means, and standard deviations to summarize the data.
 - **Diagnostic Analysis:** Assessment of data distribution and assumptions for statistical testing.
 - **Inferential Analysis:** Hypothesis testing using parametric methods (ANOVA, T-Tests, Regression).
-

Data Analysis & Results Interpretation

Stage 03 in the research

FINAL PROJECT REPORT

DETERMINANTS OF SMARTPHONE PRICINGS

A Statistical Analysis of Hardware Specifications & Brand Equity

3 – Data Analysis & Results Interpretations:

PHASE A:

Managing Variable View

Prior to analysis, the dataset was configured within the SPSS Variable View to ensure data integrity.

- **Measurement Scales:** Variables were strictly defined as **Scale** (Price, RAM, Battery, Camera), **Nominal** (Brand, Model, 5G Support), or **Ordinal** (OS).
- **Coding:** Categorical variables were automatically and numerically coded (e.g., 5G Support: 0 = No, 1 = Yes) to facilitate statistical processing.
- **Validation:** The dataset was verified for missing values and inconsistencies, ensuring a complete sample of 558 for all subsequent tests.

PHASE B:

The shape of the distribution (Data Normality check)

To ensure the validity of the statistical methods, the distribution of the continuous variables was assessed using both statistical tests and visual inspection.

Normality Testing:

The **Shapiro-Wilk** test was selected for its high sensitivity in detecting deviations from normality. As presented in the preliminary analysis, the results for all continuous variables (Price, RAM, Battery, Camera) yielded significance values of $p < .001$. **This indicates a statistically significant deviation from a perfect normal distribution.**

Case Processing Summary:

Table 3: Case Processing Summary in Normality test

Case Processing Summary						
	Valid		Cases Missing		Total	
	N	Percent	N	Percent	N	Percent
Price (\$)	558	100.0%	0	0.0%	558	100.0%
RAM (GB)	558	100.0%	0	0.0%	558	100.0%
ROM (GB)	558	100.0%	0	0.0%	558	100.0%
Camera (Megapixels)	558	100.0%	0	0.0%	558	100.0%
Battery (mAh)	558	100.0%	0	0.0%	558	100.0%
Display Size (Inch)	558	100.0%	0	0.0%	558	100.0%
Charging Support (Watt)	558	100.0%	0	0.0%	558	100.0%

Normality Tests Processing:

Table 4: Table of Normality test

Tests of Normality						
	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Price (\$)	.068	558	.000	.949	558	.000
RAM (GB)	.200	558	.000	.865	558	.000
ROM (GB)	.246	558	.000	.787	558	.000
Camera (Megapixels)	.279	558	.000	.800	558	.000
Battery (mAh)	.169	558	.000	.883	558	.000
Display Size (Inch)	.094	558	.000	.948	558	.000
Charging Support (Watt)	.206	558	.000	.868	558	.000

a. Lilliefors Significance Correction

Visual Inspection & Skewness:

Visual analysis of Histograms confirmed non-normal patterns. Specifically:

- **Price:** Exhibited a multi-modal distribution, reflecting distinct market segments (Budget, Mid-range, Premium).
- **RAM/Storage:** Showed discrete clustering due to fixed industry standards (e.g., 4GB, 8GB, 12GB) rather than a continuous curve.

Price Histogram:

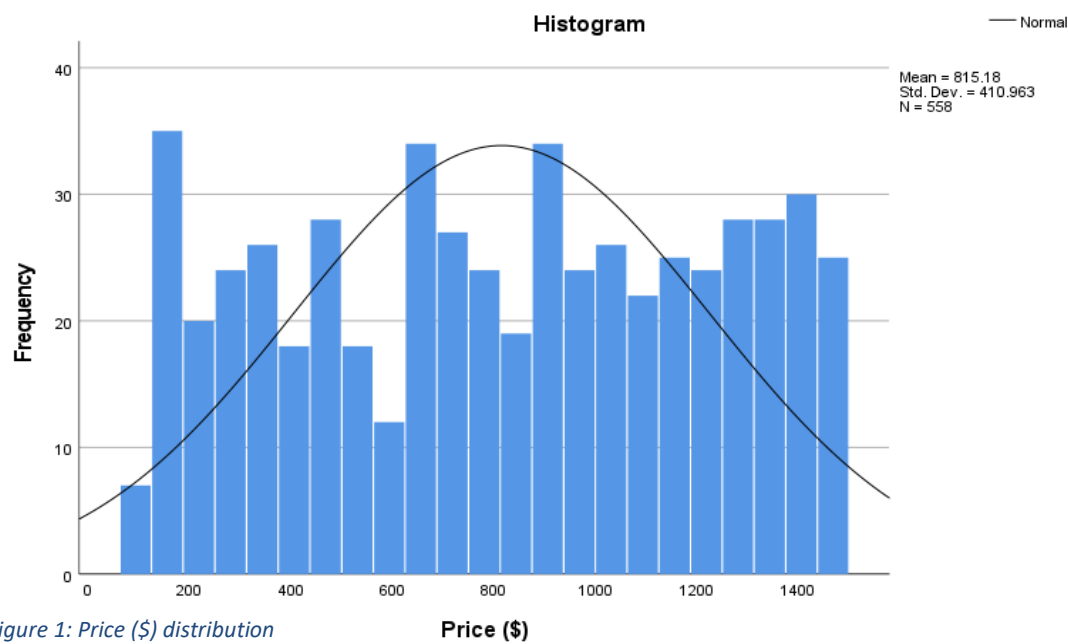


Figure 1: Price (\$) distribution

RAM/Storage Histogram:

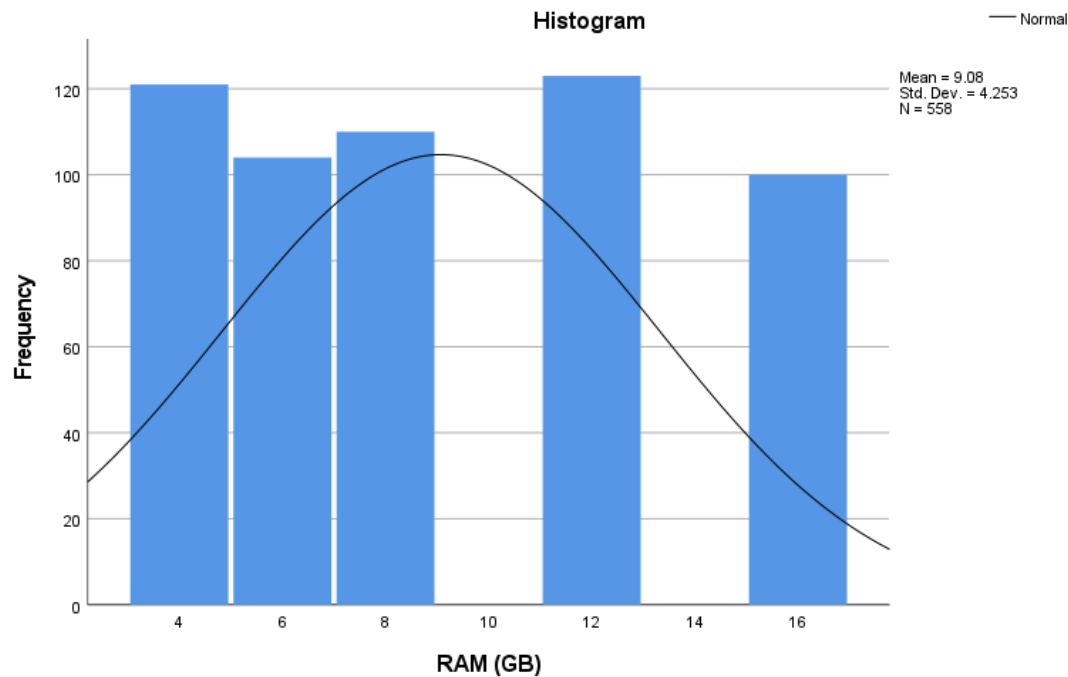


Figure 2: RAM Distribution

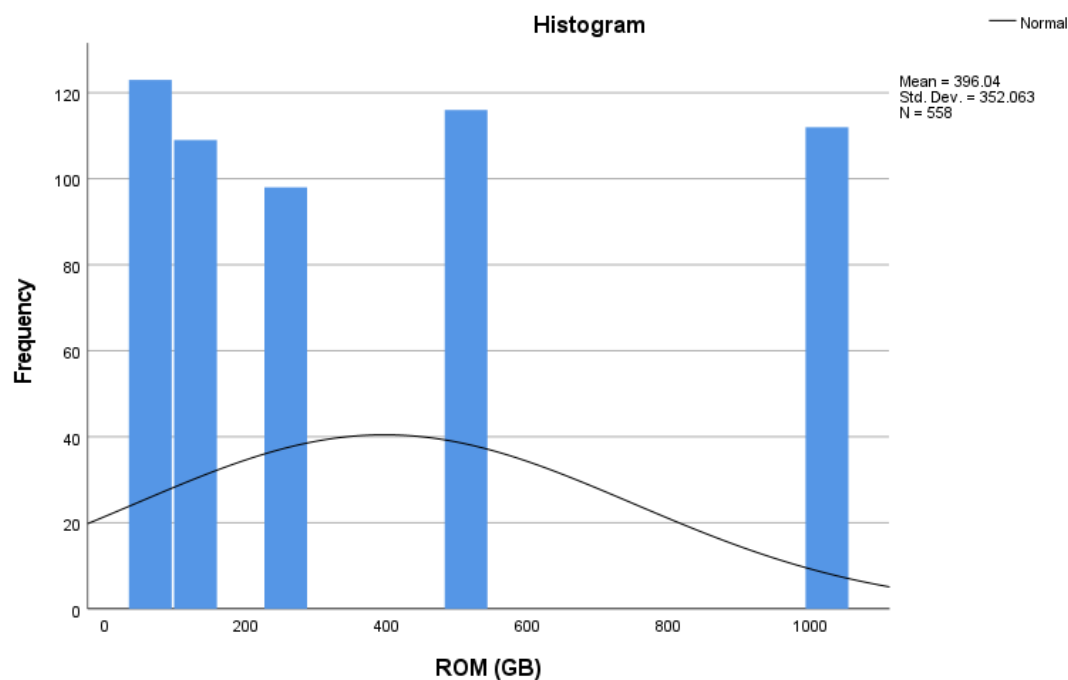


Figure 3: ROM Distribution

Selection of Statistical Methods: (defence against using abnormal data):

Despite the Shapiro-Wilk results indicating non-normality, this study proceeds with **Parametric Tests**. This decision is grounded in the **Central Limit Theorem (CLT)**, which implies that for large sample sizes ($N > 30$), the sampling distribution of the mean approximates normality regardless of the shape of the population distribution.

Given the robust sample size of **N = 558**, the risk of statistical error due to non-normality is negligible. Therefore, parametric methods were chosen to ensure methodological consistency with the Linear Regression analysis (Objective 5) and to maximize statistical power.

PHASE C:

Descriptive statistics:

This phase provides a comprehensive summary of the dataset, detailing the frequency distributions of categorical variables and the central tendencies of continuous technical specifications. This analysis addresses **Objective 6**, offering insights into market segmentation and feature patterns.

Categorical Distribution:

The frequency analysis highlights the market composition of the sample (\$N=558\$). As detailed in the following table, the dataset represents a competitive landscape among five major manufacturers.

- **Market Share: Google & OnePlus** accounts for the largest share of the sample with **116** models (20.8%), followed closely by **Apple Inc. (19.2%)**. This indicates a highly competitive mid-range market presence.
- **Technology Adoption:** In terms of connectivity, **46.8%** of the devices in the sample are 5G-enabled, while **53.2%** remain limited to 4G/LTE. This suggests that while 5G is becoming standard, legacy of 4G models still retain a significant market portion.

Frequency Distribution of Brands and Connectivity:

Table 5: Frequency of Brands Distribution

Mobile Brand					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Apple	107	19.2	19.2	19.2
	Google	116	20.8	20.8	40.0
	OnePlus	116	20.8	20.8	60.8
	Samsung	105	18.8	18.8	79.6
	Xiaomi	114	20.4	20.4	100.0
	Total	558	100.0	100.0	

Table 6: Connectivity Distribution Frequency

5g Support					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	297	53.2	53.2	53.2
	Yes	261	46.8	46.8	100.0
	Total	558	100.0	100.0	

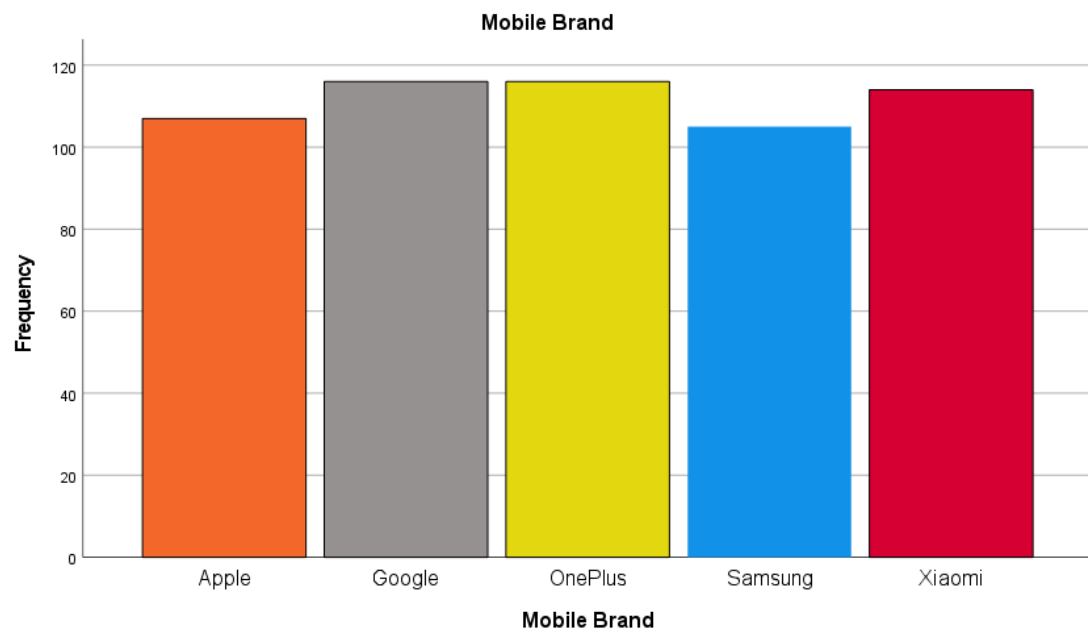


Figure 4: Mobile Brands Frequency Distribution

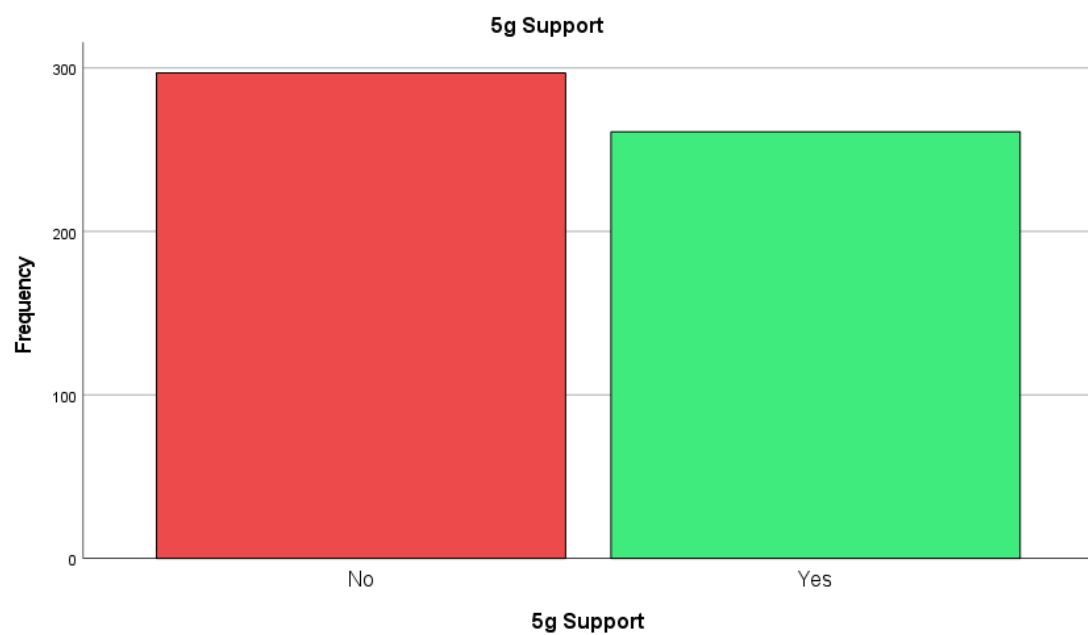


Figure 5: Connectivity frequency Distribution

Descriptive statistics of Technical Specification:

The following Table presents the descriptive statistics (Mean, Standard Deviation, Minimum, Maximum) for the key continuous variables. These metrics provide the baseline for understanding the "Standard" smartphone features in the current market:

Descriptive Statistics											
	N Statistic	Range Statistic	Minimum Statistic	Maximum Statistic	Mean Statistic	Std. Deviation Statistic	Variance Statistic	Skewness		Kurtosis	
								Statistic	Std. Error	Statistic	Std. Error
Price (\$)	558	1393	105	1498	815.18	410.963	168890.536	-.068	.103	-1.222	.206
RAM (GB)	558	12	4	16	9.08	4.253	18.088	.405	.103	-1.193	.206
ROM (GB)	558	960	64	1024	396.04	352.063	123948.063	.869	.103	-.709	.206
Camera (Megapixels)	558	188	12	200	82.57	60.782	3694.458	.997	.103	-.235	.206
Battery (mAh)	558	2000	4000	6000	5016.13	717.009	514102.044	-.035	.103	-1.340	.206
Display Size (Inch)	558	1.7	5.5	7.2	6.365	.5056	.256	-.031	.103	-1.229	.206
Charging Support (Watt)	558	102	18	120	63.50	36.384	1323.816	.371	.103	-1.333	.206
Consumer Rating	558	1.5	3.5	5.0	4.220	.4409	.194	.114	.103	-1.173	.206
Valid N (listwise)	558										

Table 7: Descriptive Stats: Technical Specs

Interpretation of Descriptive findings.

- Price Variability:** The average price of a mobile phone in this dataset is **815 \$**, but the high standard deviation (410.96) confirms massive price disparity. This supports the existence of three distinct pricing tiers: Budget (~ 105 \$ to 815 \$), Mid-Range (815\$), and Premium (1200 \$).
- Battery Standardization:** The mean battery capacity is approximately **5016 mAh** with a relatively lower standard deviation compared to price. This suggests that high-capacity batteries have become an industry standard across all price points, rather than a premium feature.
- Camera & Memory Diversity:** RAM and Camera specifications show significant variance, indicating these are key differentiators used by brands to distinguish "Pro" models from base models.

Addressing Objective 06:

Based on the descriptive data, clear feature patterns emerge that distinguish market segments:

- Low-Priced Models:** Typically characterized by entry-level specifications, specifically **4GB RAM**, smaller storage capacities starting at **64GB**, and prices dipping as low as **\$105**.
- Mid-Range Models:** Cluster around the mean price of **\$815.18**, offering standard features like **8GB to 10GB RAM** (Mean = 9.08GB) and standardized **5000mAh** batteries (Mean = 5016 mAh).

- **High-Priced Models:** Defined by "Max" values in the dataset, specifically **16GB RAM**, massive **1TB (1024GB)** storage options, and premium display sizes peaking at **7.2 inches**, commanding prices up to **\$1498**.

PHASE D:

Inferential Statistics:

This phase addresses the core research objectives through hypothesis testing. The level of significance was set at 0.05 for all inferential tests.

Hypothesis 01: Brand Price Differentiation:

Step 1: Define Hypothesis

- **Null Hypothesis (H₀):** There is no significant difference in the mean price across different mobile phone brand).
- **Alternative Hypothesis (H₁):** At least one brand mean is different, means there is a statistically significant difference in the mean price across different mobile phone brands.

Step 2: Level of Significance

- alpha = 0.05

Step 3: Test Statistic

- **Test Used:** One-Way ANOVA
- **F-statistic:** F = 0.188

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
Price (\$)	Based on Mean	.679	4	553	.607
	Based on Median	.635	4	553	.638
	Based on Median and with adjusted df	.635	4	549.394	.638
	Based on trimmed mean	.675	4	553	.609

Table 8: Levene's Test for Price between brands

ANOVA

Price (\$)

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	127638.543	4	31909.636	.188	.945
Within Groups	93944390.25	553	169881.357		
Total	94072028.79	557			

Table 9: Table of ANOVA for Prices (\$)

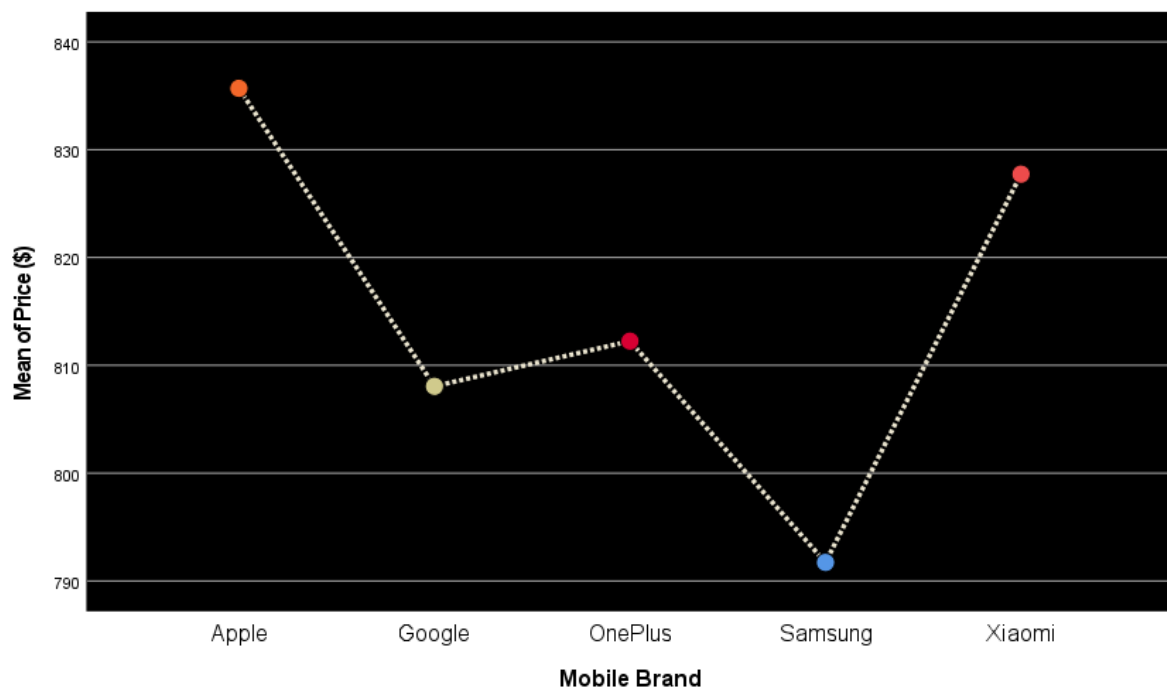


Figure 6: Mean Price (\$) line plot

Step 4: P-Value (Decision Criteria)

- **Sig. value:** $p = 0.945$
- **Criteria:**
- If $p \leq 0.05$, Reject H_0 .
- If $p > 0.05$, Fail to Reject H_0 .

Step 5: Make Decision

- Since the p-value (0.945) is greater than the significance level:
- we **Fail to Reject the Null Hypothesis**.

Step 6: Conclusion

There is insufficient statistical evidence to claim that the average price varies significantly between brands. The data suggests that all five brands (Samsung, Apple, Xiaomi, OnePlus, Google) have statistically similar mean prices in this market sample.

Hypothesis 02: Correlation of Specification with prices:

Step 1: Define Hypothesis

- **Null Hypothesis (H_0):** There is no significant relationship between hardware specifications and Price.
- **Alternative Hypothesis (H_1):** There is a significant linear relationship between hardware specifications and Price.

Step 2: Level of Significance

- $\alpha = 0.05$

Step 3: Test Statistic

- **Test Used:** Pearson Product-Moment Correlation (r).
- **Observed Coefficients (r):**
 - Price vs. RAM: $r = -.005$
 - Price vs. Camera: $r = .030$
 - Price vs. Battery: $r = -.010$
 - Price vs. Display: $r = -.063$

		Correlations						
		Price (\$)	RAM (GB)	ROM (GB)	Camera (Megapixels)	Battery (mAh)	Display Size (Inch)	Charging Support (Watt)
Price (\$)	Pearson Correlation	1	-.005	-.034	.030	-.010	-.063	.002
	Sig. (2-tailed)		.906	.429	.475	.812	.139	.960
	N	558	558	558	558	558	558	558
RAM (GB)	Pearson Correlation	-.005	1	.030	-.045	.038	.022	-.009
	Sig. (2-tailed)	.906		.477	.289	.365	.608	.832
	N	558	558	558	558	558	558	558
ROM (GB)	Pearson Correlation	-.034	.030	1	.039	-.043	-.008	.027
	Sig. (2-tailed)	.429	.477		.356	.315	.853	.523
	N	558	558	558	558	558	558	558
Camera (Megapixels)	Pearson Correlation	.030	-.045	.039	1	.032	-.112**	.035
	Sig. (2-tailed)	.475	.289	.356		.445	.008	.403
	N	558	558	558	558	558	558	558
Battery (mAh)	Pearson Correlation	-.010	.038	-.043	.032	1	.007	.009
	Sig. (2-tailed)	.812	.365	.315	.445		.878	.830
	N	558	558	558	558	558	558	558
Display Size (Inch)	Pearson Correlation	-.063	.022	-.008	-.112**	.007	1	.060
	Sig. (2-tailed)	.139	.608	.853	.008	.878		.156
	N	558	558	558	558	558	558	558
Charging Support (Watt)	Pearson Correlation	.002	-.009	.027	.035	.009	.060	1
	Sig. (2-tailed)	.960	.832	.523	.403	.830	.156	
	N	558	558	558	558	558	558	558

** . Correlation is significant at the 0.01 level (2-tailed).

Table 10: Table of Correlation

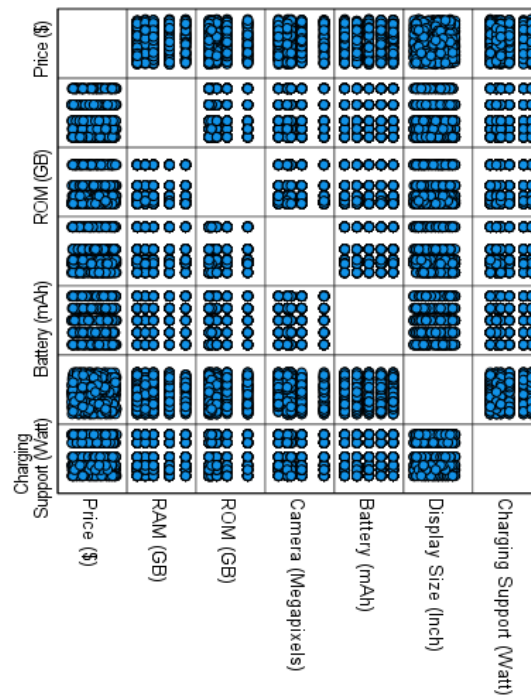


Figure 7: Correlation Scatter Graph

Step 4: P-Value (Decision Criteria)

- **Sig. (2-tailed) values:**
 - Price vs. RAM: $p = .906$
 - Price vs. Camera: $p = .475$
 - Price vs. Battery: $p = .812$
 - Price vs. Display: $p = .139$
- **Criteria:** Since all calculated p-values are > 0.05 , then the results are not statistically significant.

Step 5: Make Decision

We **Fail to Reject the Null Hypothesis** for all tested variables.

Step 6: Conclusion

Statistical analysis reveals that there is **no significant linear relationship** between the technical specifications and the retail price in this dataset. The Pearson correlation coefficients are negligible (near zero), indicating that an increase in RAM, Battery size, or Camera megapixels does not statistically correspond to an increase in Price for this specific sample of devices.

Hypothesis 03: Impact of 5G Connectivity on the Price:

Step 1: Define Hypothesis

- **Null Hypothesis (H_0):** There is no significant difference in the mean price between 5G supported and non-5G devices.
- **Alternative Hypothesis (H_1):** The mean price of 5G supported phones is statistically significantly different from that of non-5G phones.

Step 2: Level of Significance

- $\alpha = 0.05$

Step 3: Test Statistic

- **Test Used:** Independent Samples T-Test.
- **Assumption Check:** Levene's Test for Equality of Variances yielded $p = .0349$ ($> .05$), so equal variances were assumed.
- **t-statistic:** $t\text{-stat} = 1.095$

Group Statistics					
5g Support		N	Mean	Std. Deviation	Std. Error Mean
Price (\$)	No	297	833.04	402.052	23.329
	Yes	261	794.85	420.724	26.042

Table 11: Table of Groups inside connectivity sample

Independent Samples Test									
Levene's Test for Equality of Variances				t-test for Equality of Means					
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference Lower Upper
Price (\$)	Equal variances assumed	.878	.349	1.095	556	.274	38.186	34.861	-30.289 106.662
	Equal variances not assumed			1.092	539.538	.275	38.186	34.964	-30.495 106.868

Table 12: Independent Sample Test

Step 4: P-Value (Decision Criteria)

- **Sig. (2-tailed) value:** $p = 0.274$
- **Criteria:** Since the calculated p-value (0.274) is greater than alpha, the result is not statistically significant.

Step 5: Make Decision

We **Fail to Reject the Null Hypothesis**.

Step 6: Conclusion

The statistical evidence suggests that 5G connectivity is not a significant determinant of price in this dataset. The mean price of non-5G phones (\$833.04) was statistically indistinguishable from 5G-enabled phones (\$794.85). This implies that older 4G flagship models and newer 5G mid-range models are competing at similar price points, neutralizing any "5G Premium."

Hypothesis 04: Consumer Rating Analysis:

Step 1: Define Hypothesis

- **Null Hypothesis (H₀):** The average consumer rating for mobile phones is statistically equal to the market benchmark of 4.0.
- **Alternative Hypothesis (H₁):** The average consumer rating for mobile phones is significantly different from the market benchmark of 4.0.

Step 2: Level of Significance

- alpha = 0.05

Step 3: Test Statistic

- **Test Used:** One-Sample T-Test
- **Test Value:** 4.0
- **t-statistic:** 11.809
- **Degrees of Freedom:** 557

One-Sample Statistics				
	N	Mean	Std. Deviation	Std. Error Mean
Consumer Rating	558	4.220	.4409	.0187

Table 13: Table of One Sample Statistics

One-Sample Test						
Test Value = 4						
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
Consumer Rating	11.809	557	<.001	.2204	.184	.257

Table 14: One Sample Test

Step 4: P-Value (Decision Criteria)

- **Sig. (2-tailed) value:** Less than .001 (shown as < .001 in SPSS).

- Criteria:** Since the calculated p-value is less than 0.05, the result is statistically significant.

Step 5: Make Decision

We **Reject the Null Hypothesis**.

Step 6: Conclusion

*There is sufficient statistical evidence to conclude that the average consumer rating is significantly different from the market benchmark of 4.0. The sample mean rating was **4.22**, which is significantly higher than the benchmark, indicating that consumer satisfaction in this market is generally positive and exceeds the neutral standard.*

Hypothesis 05: Predictive Price Modelling:

Step 1: Define Hypothesis

- Null Hypothesis (H₀):** The technical specifications do not significantly predict the prices of the mobile phones.
- Alternative Hypothesis (H₁):** The technical specifications can significantly predict the prices of the mobile phones.

Step 2: Level of Significance

- alpha = 0.05

Step 3: Test Statistic

- Test Used:** Multiple Linear Regression
- Model Summary:** R Square = .006. (This means the model explains only 0.6% of the price variation)
- F-statistic (ANOVA):** .543

Model Summary ^b				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.077 ^a	.006	-.005	411.978

a. Predictors: (Constant), Charging Support (Watt), RAM (GB), Battery (mAh), ROM (GB), Display Size (Inch), Camera (Megapixels)

b. Dependent Variable: Price (\$)

Table 15: Regression Model Summary

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	553084.453	6	92180.742	.543	.775 ^b
	Residual	93518944.34	551	169725.852		
	Total	94072028.79	557			

a. Dependent Variable: Price (\$)

b. Predictors: (Constant), Charging Support (Watt), RAM (GB), Battery (mAh), ROM (GB), Display Size (Inch), Camera (Megapixels)

Table 16: Table of ANOVA for regression model

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	1161.443	258.454		4.494	<.001		
	RAM (GB)	-.095	4.115	-.001	-.023	.982	.995	1.005
	ROM (GB)	-.042	.050	-.036	-.837	.403	.995	1.005
	Camera (Megapixels)	.169	.290	.025	.584	.559	.981	1.019
	Battery (mAh)	-.007	.024	-.012	-.283	.777	.995	1.005
	Display Size (Inch)	-49.111	34.824	-.060	-1.410	.159	.983	1.017
	Charging Support (Watt)	.067	.481	.006	.139	.889	.994	1.006

a. Dependent Variable: Price (\$)

Table 17: Table of Coefficients for regression model

Residuals Statistics^a

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	740.12	897.07	815.18	31.511	558
Residual	-741.222	745.909	.000	409.753	558
Std. Predicted Value	-2.382	2.599	.000	1.000	558
Std. Residual	-1.799	1.811	.000	.995	558

a. Dependent Variable: Price (\$)

Table 18: The residuals of regression model

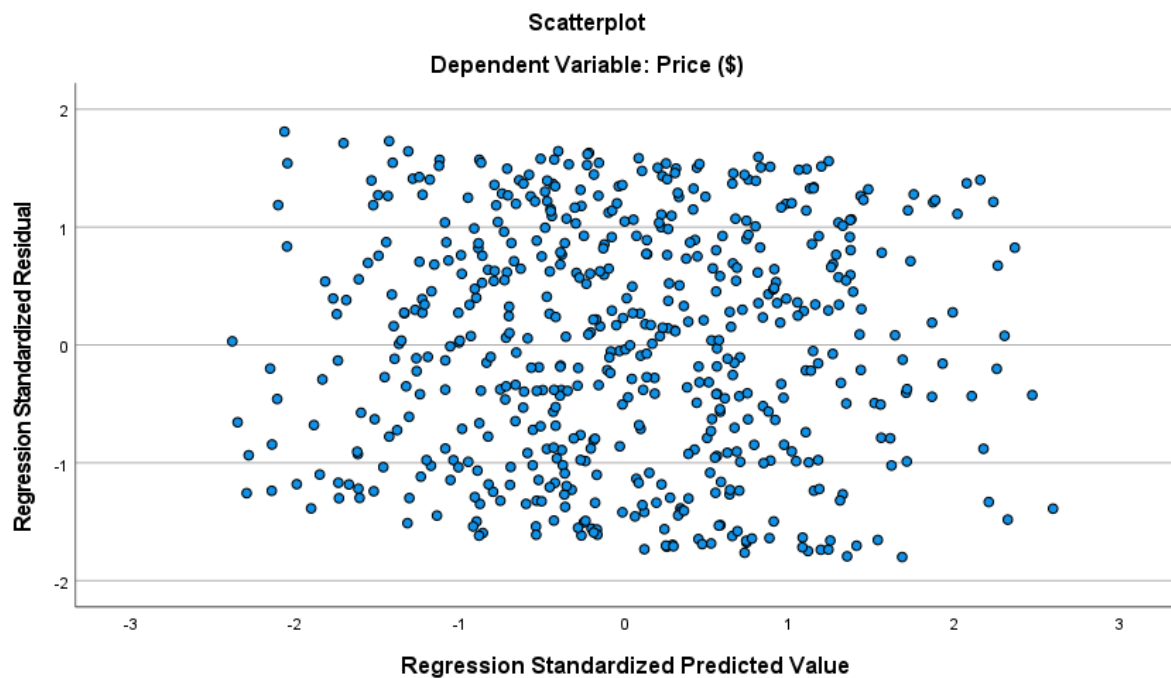


Figure 8: Regression Model Scatter Plot

Step 4: P-Value (Decision Criteria)

- **Sig. value (ANOVA Table):** .775.
- **Criteria:** Since the calculated p-value (.775) is greater than 0.05, the regression model is not statistically valid.

Step 5: Make Decision

We **Fail to Reject the Null Hypothesis**.

Step 6: Conclusion

*The regression analysis indicates that the selected technical specifications are **not** significant predictors of mobile phone prices in this dataset. The overall model fit was poor ($F = .543$, $p = .775$), and **none** of the individual predictors (RAM, Battery, Camera, etc.) showed statistical significance (e.g., RAM $p = .982$, Camera $p = .559$). This **suggests that price is likely determined by factors outside of these hardware specifications, such as brand marketing, build quality, or software ecosystem, rather than raw performance metrics.***

The Conclusion & Limitations of the Study

Stage 04 in the research

FINAL PROJECT REPORT

DETERMINANTS OF SMARTPHONE PRICINGS

A Statistical Analysis of Hardware Specifications & Brand Equity

4 – Conclusion:

Summary:

This research aimed to deconstruct the pricing determinants of the modern smartphone market by analysing 558 devices across five major brands. The study sought to establish a statistical link between tangible hardware specifications (RAM, Battery, Camera) and retail price.

General Findings & Justification:

Contrary to initial expectations, the inferential analysis revealed that **neither brand identity nor technical specifications** are significant predictors of price in this dataset.

The One-Way ANOVA showed no significant price disparity among brands ($p=.945$), justified by the diverse product portfolios of manufacturers like Samsung and Apple, which span from budget to ultra-premium tiers, equalizing the average costs.

Furthermore, the *Linear Regression model failed to predict price based on hardware* ($R^2 = .006$), suggesting that ***market value is likely driven by intangible factors, such as brand equity, software ecosystem, and build quality, rather than raw performance metrics like RAM or battery size.***

Additionally, the analysis confirmed that 5G connectivity is no longer a premium differentiator ($p=.274$), reflecting its commoditization in the mid-range market.

Limitations of the study:

1. **Data Distribution:** The Shapiro-Wilk test indicated significant non-normality ($p<.001$) in all variables, necessitating the use of the Central Limit Theorem to justify parametric testing.
 2. **Variable Scope:** The dataset focused strictly on hardware. The exclusion of "soft" variables (e.g., screen technology type, processor generation, build materials) limited the regression model's ability to explain price variance.
 3. **Secondary Data Constraints:** Reliance on aggregated secondary data may have introduced high variance, as evidenced by the extreme standard deviations observed in the descriptive analysis.
-

References

Stage 05 in the research

FINAL PROJECT REPORT

DETERMINANTS OF SMARTPHONE PRICINGS

A Statistical Analysis of Hardware Specifications & Brand Equity

References

Field, A. (2013). *Discovering Statistics Using IBM SPSS Statistics*. London: Sage Publications.

IBM Corp. (2021). IBM Corp.

Kaggle Data Repository. (n.d.). *Mobile Phone Specifications and Prices Dataset*. Retrieved from <https://www.kaggle.com/datasets/shahzadi786/world-smartphone-market-2025>

Appendix

The Backstage Area of the Report

FINAL PROJECT REPORT

DETERMINANTS OF SMARTPHONE PRICINGS

A Statistical Analysis of Hardware Specifications & Brand Equity

TurnitIn Originality Report:

Justice Gul Muhammad Library – Fast NU, Lahore.

FINAL PROJECT REPORT

DETERMINANTS OF SMARTPHONE PRICINGS

A Statistical Analysis of Hardware Specifications & Brand Equity

>Warning # 849 in column 23. Text: en_PK
>The LOCALE subcommand of the SET command has an invalid parameter. It could
>not be mapped to a valid backend locale.

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  /READNAMES=ON
  /DATATYPEMIN PERCENTAGE=95.0
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DATASET ACTIVATE DataSet1.

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```

DATASET ACTIVATE DataSet1.

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```

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  /INTO Mobile_brand_code
  /PRINT.
```

brand into Mobile_brand_code (Mobile Brand)
Old Value New Value Value Label

Apple	1	Apple
Google	2	Google
OnePlus	3	OnePlus
Samsung	4	Samsung
Xiaomi	5	Xiaomi

```

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  /PRINT.
five_g_support into support_5g (5g Support)
Old Value  New Value  Value Label

```

No	1	No
Yes	2	Yes

```

AUTORECODE VARIABLES=os
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  /PRINT.
os into OS_Code (Operating system)
Old Value  New Value  Value Label

```

Android	1	Android
iOS	2	iOS

```

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rging_watt
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  /CINTERVAL 95
  /MISSING LISTWISE
  /NOTOTAL.

```

Explore

Notes

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	Cases Used	Statistics are based on cases with no missing values for any dependent variable or factor used.
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[DataSet1] E:\BS Business Analytics\Fall 2025\Data Analysis 02\Semester Project\Final Folder\Project File.sav

Case Processing Summary

	Valid		Cases Missing		Total	
	N	Percent	N	Percent	N	Percent
Price (\$)	558	100.0%	0	0.0%	558	100.0%
RAM (GB)	558	100.0%	0	0.0%	558	100.0%
ROM (GB)	558	100.0%	0	0.0%	558	100.0%
Camera (Megapixels)	558	100.0%	0	0.0%	558	100.0%
Battery (mAh)	558	100.0%	0	0.0%	558	100.0%
Display Size (Inch)	558	100.0%	0	0.0%	558	100.0%
Charging Support (Watt)	558	100.0%	0	0.0%	558	100.0%

Descriptives

			Statistic	Std. Error
Price (\$)	Mean		815.18	17.397
	95% Confidence Interval for Mean	Lower Bound	781.00	
		Upper Bound	849.35	
	5% Trimmed Mean		816.90	
	Median		822.50	
	Variance		168890.536	
	Std. Deviation		410.963	
	Minimum		105	
	Maximum		1498	
	Range		1393	
	Interquartile Range		725	
	Skewness		-.068	.103
	Kurtosis		-1.222	.206
RAM (GB)	Mean		9.08	.180
	95% Confidence Interval for Mean	Lower Bound	8.72	
		Upper Bound	9.43	
	5% Trimmed Mean		8.97	
	Median		8.00	
	Variance		18.088	
	Std. Deviation		4.253	
	Minimum		4	
	Maximum		16	
	Range		12	

Descriptives

		Statistic	Std. Error
	Interquartile Range	6	
	Skewness	.405	.103
	Kurtosis	-1.193	.206
ROM (GB)	Mean	396.04	14.904
	95% Confidence Interval for Mean	Lower Bound	366.77
		Upper Bound	425.32
	5% Trimmed Mean	379.60	
	Median	256.00	
	Variance	123948.063	
	Std. Deviation	352.063	
	Minimum	64	
	Maximum	1024	
	Range	960	
	Interquartile Range	384	
	Skewness	.869	.103
	Kurtosis	-.709	.206
Camera (Megapixels)	Mean	82.57	2.573
	95% Confidence Interval for Mean	Lower Bound	77.51
		Upper Bound	87.62
	5% Trimmed Mean	79.96	
	Median	64.00	
	Variance	3694.458	
	Std. Deviation	60.782	
	Minimum	12	
	Maximum	200	
	Range	188	
	Interquartile Range	60	
	Skewness	.997	.103
	Kurtosis	-.235	.206
Battery (mAh)	Mean	5016.13	30.353
	95% Confidence Interval for Mean	Lower Bound	4956.51
		Upper Bound	5075.75
	5% Trimmed Mean	5017.92	
	Median	5000.00	
	Variance	514102.044	
	Std. Deviation	717.009	

Descriptives

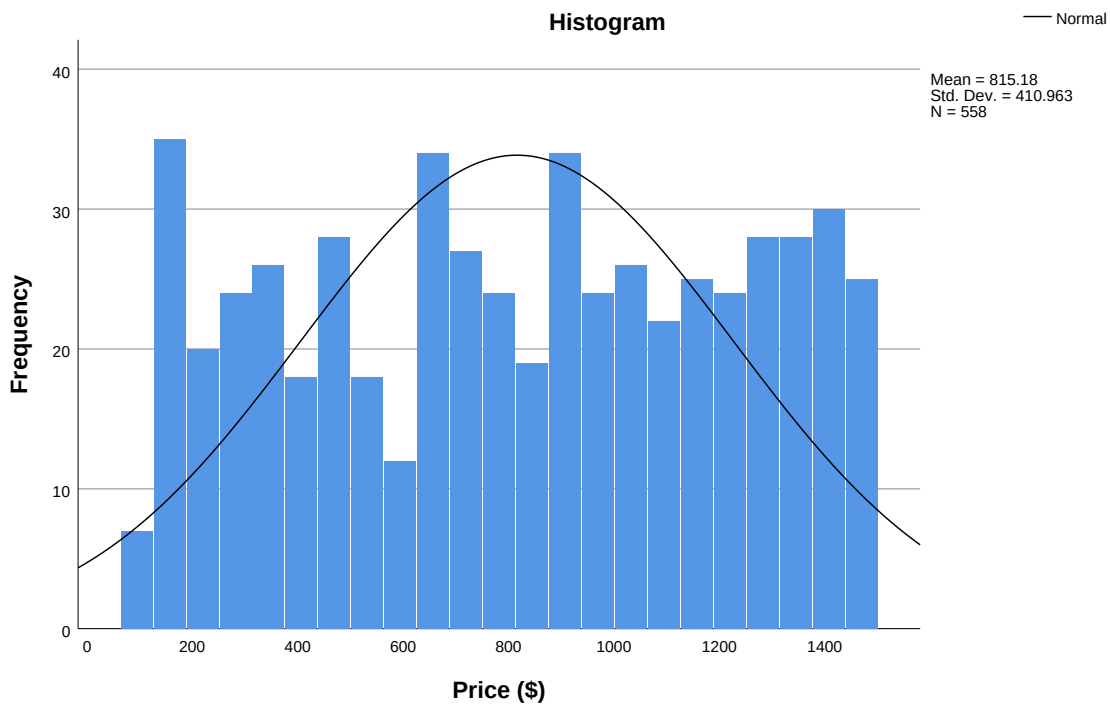
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	Range	2000	
	Interquartile Range	1000	
	Skewness	-.035	.103
	Kurtosis	-1.340	.206
Display Size (Inch)	Mean	6.365	.0214
	95% Confidence Interval for Mean	Lower Bound	6.322
		Upper Bound	6.407
	5% Trimmed Mean	6.366	
	Median	6.400	
	Variance	.256	
	Std. Deviation	.5056	
	Minimum	5.5	
	Maximum	7.2	
	Range	1.7	
	Interquartile Range	.9	
	Skewness	-.031	.103
	Kurtosis	-1.229	.206
Charging Support (Watt)	Mean	63.50	1.540
	95% Confidence Interval for Mean	Lower Bound	60.47
		Upper Bound	66.52
	5% Trimmed Mean	62.88	
	Median	44.00	
	Variance	1323.816	
	Std. Deviation	36.384	
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	Interquartile Range	67	
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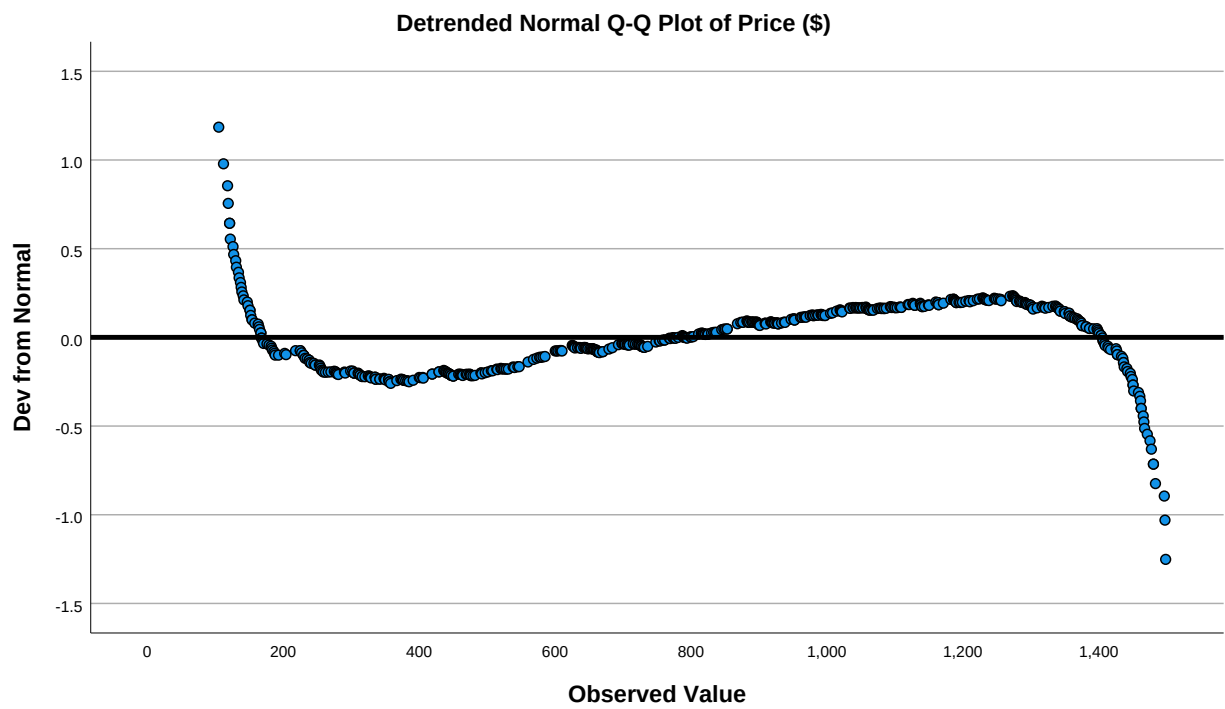
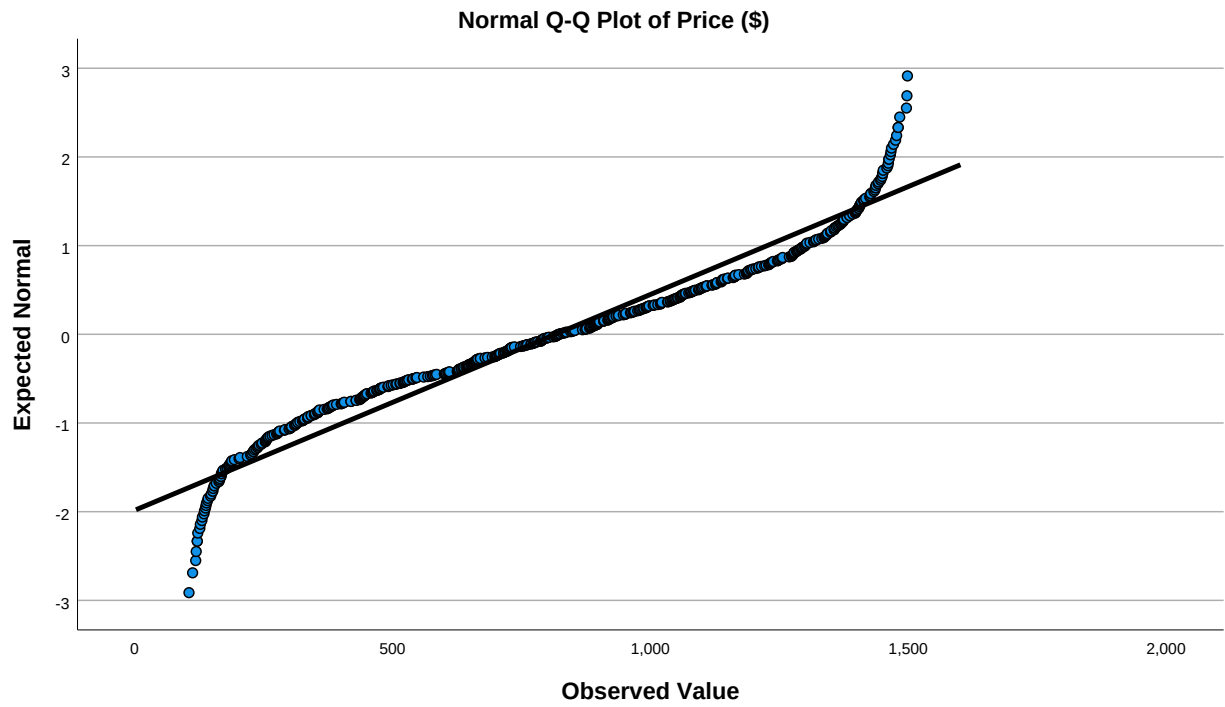
Tests of Normality

	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
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RAM (GB)	.200	558	.000	.865	558	.000
ROM (GB)	.246	558	.000	.787	558	.000
Camera (Megapixels)	.279	558	.000	.800	558	.000
Battery (mAh)	.169	558	.000	.883	558	.000
Display Size (Inch)	.094	558	.000	.948	558	.000
Charging Support (Watt)	.206	558	.000	.868	558	.000

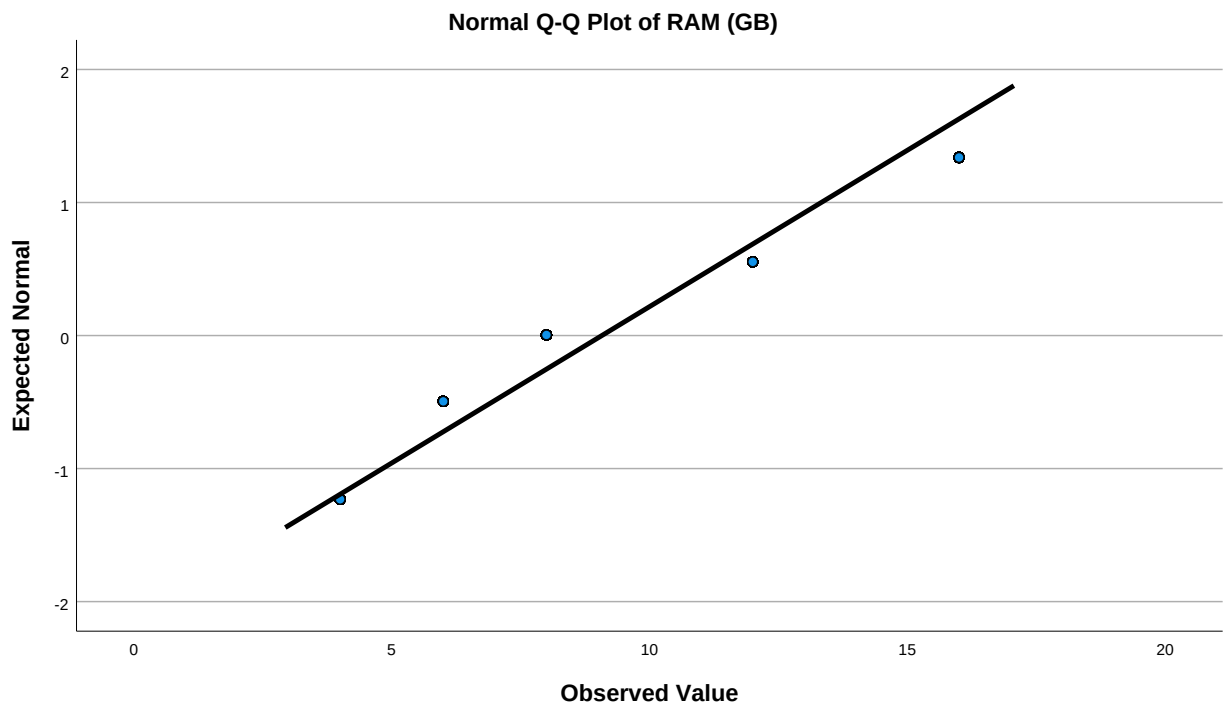
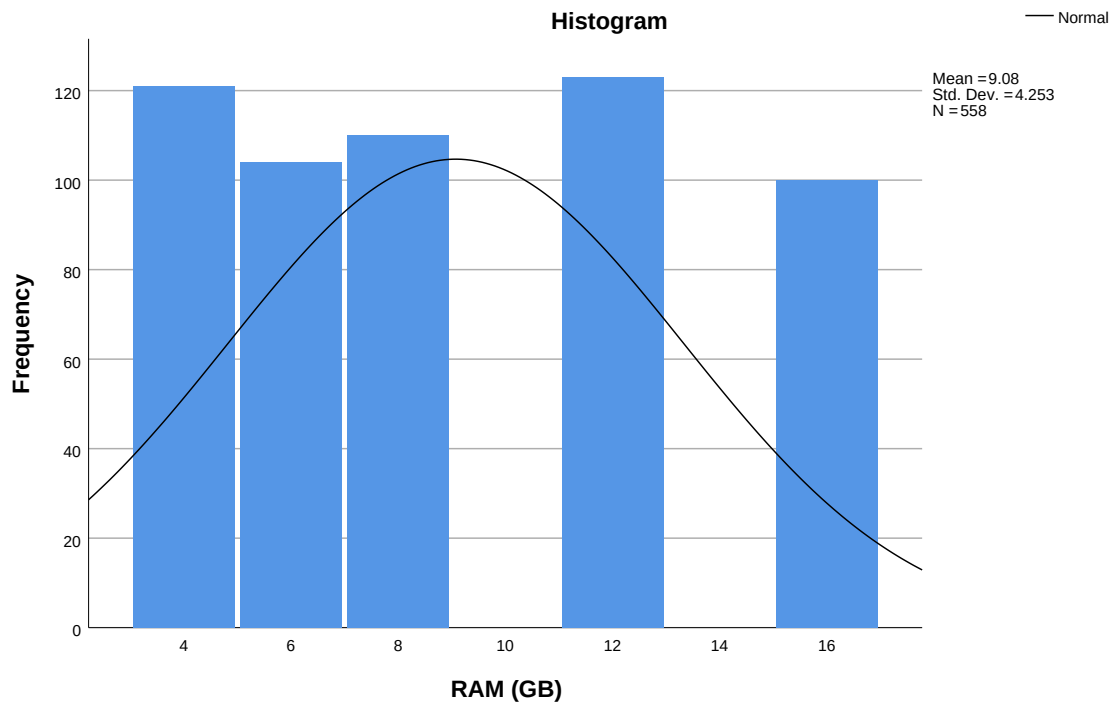
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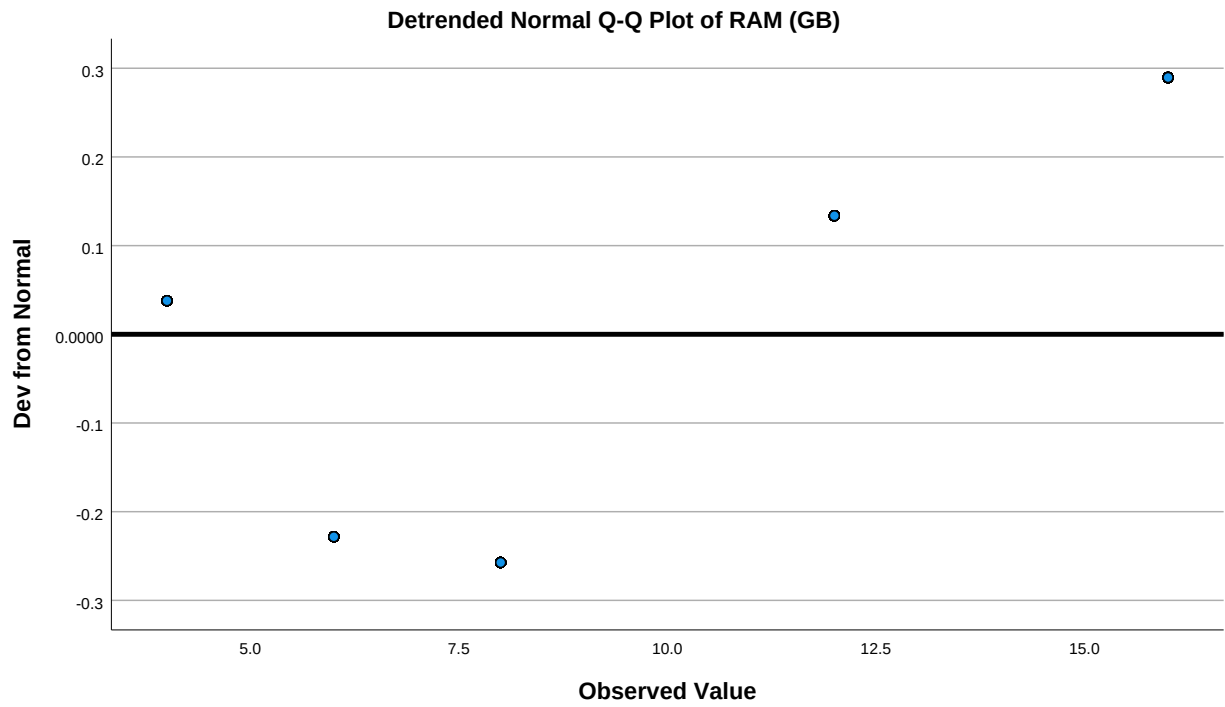
Price (\$)



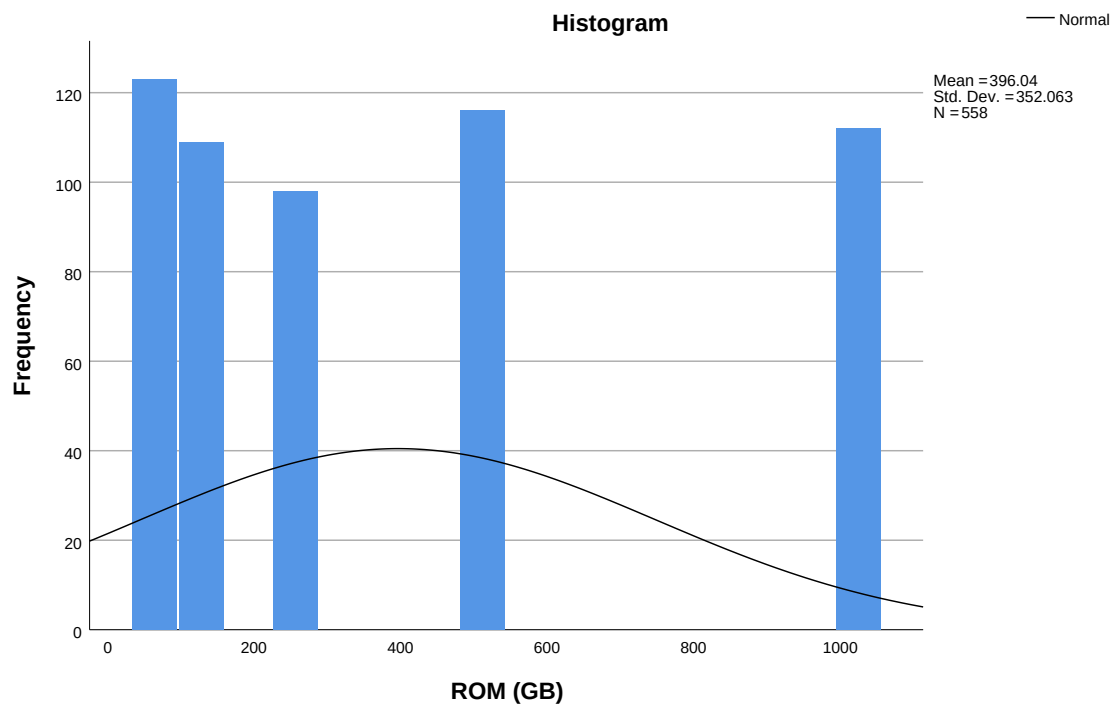


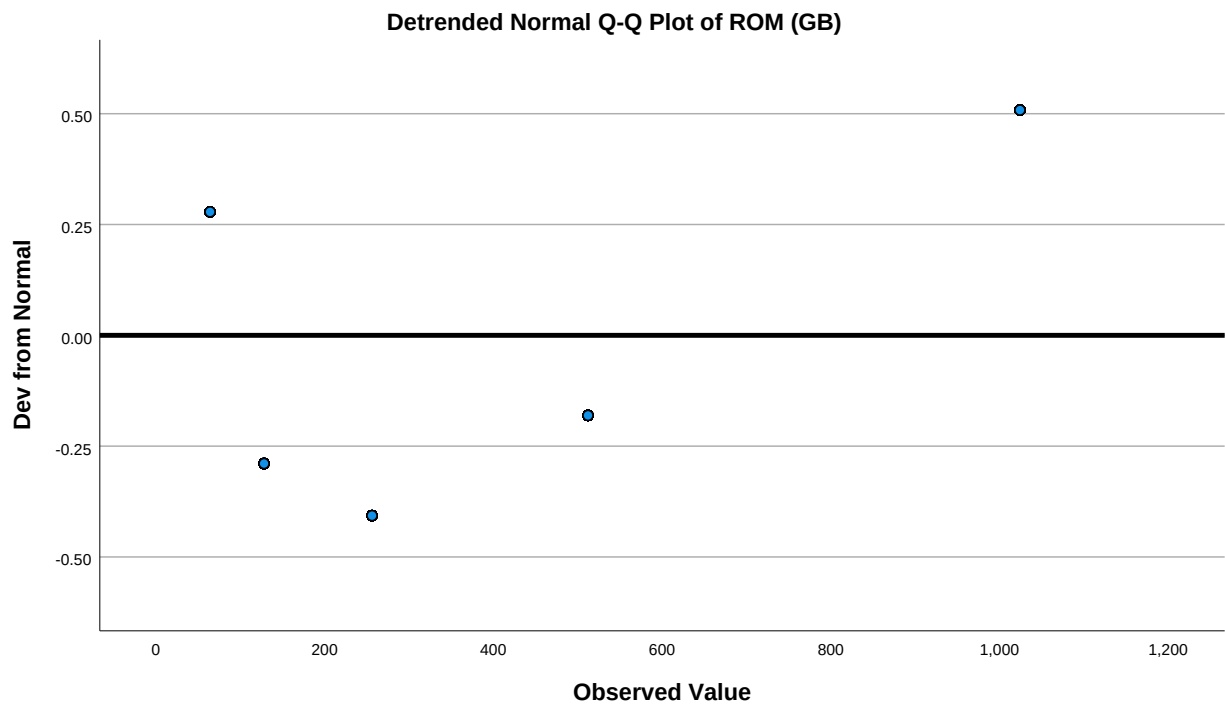
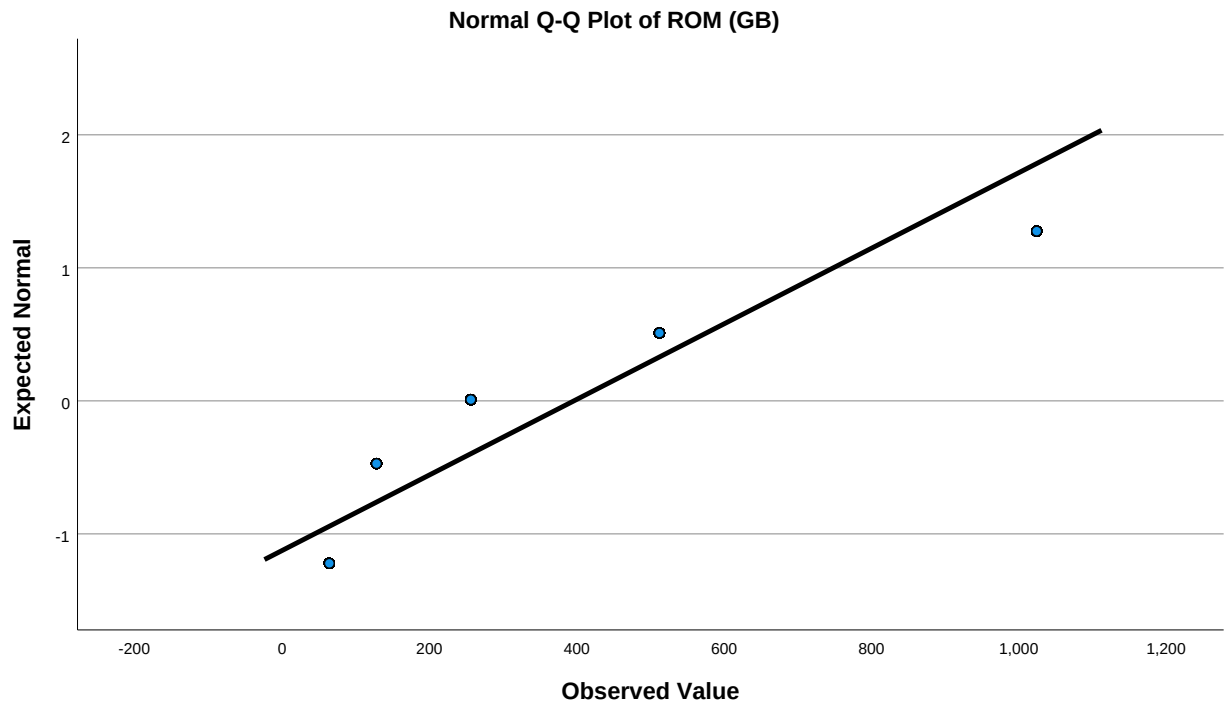
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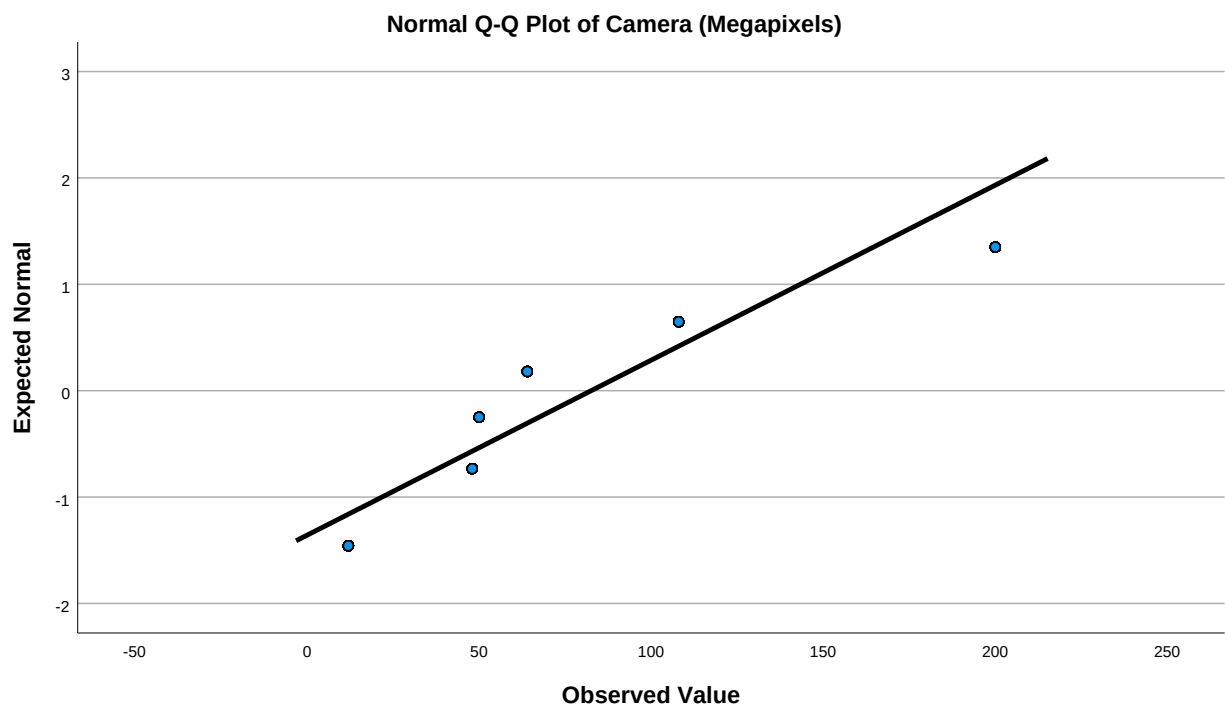
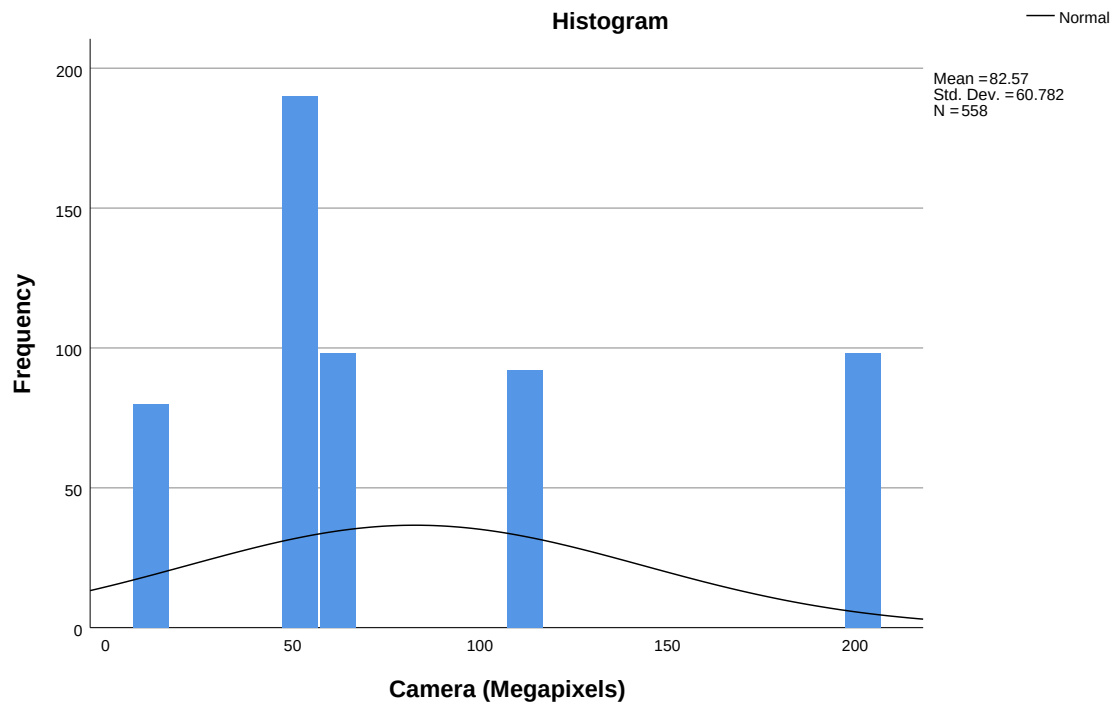


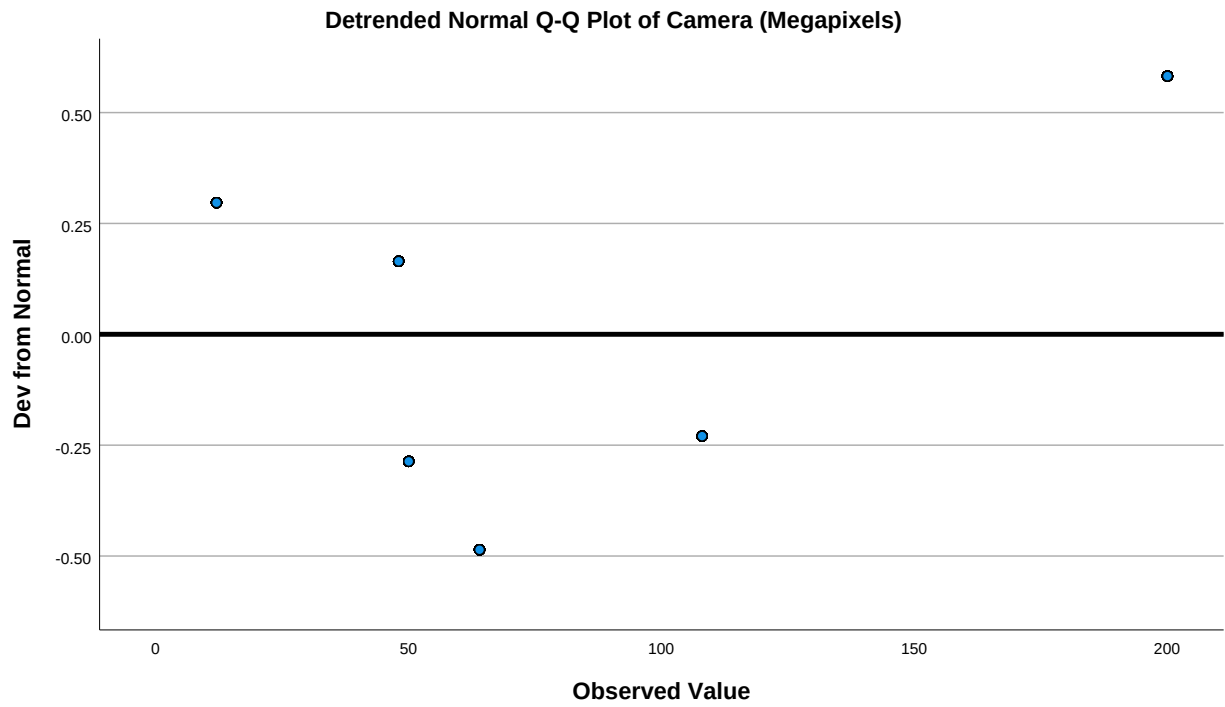
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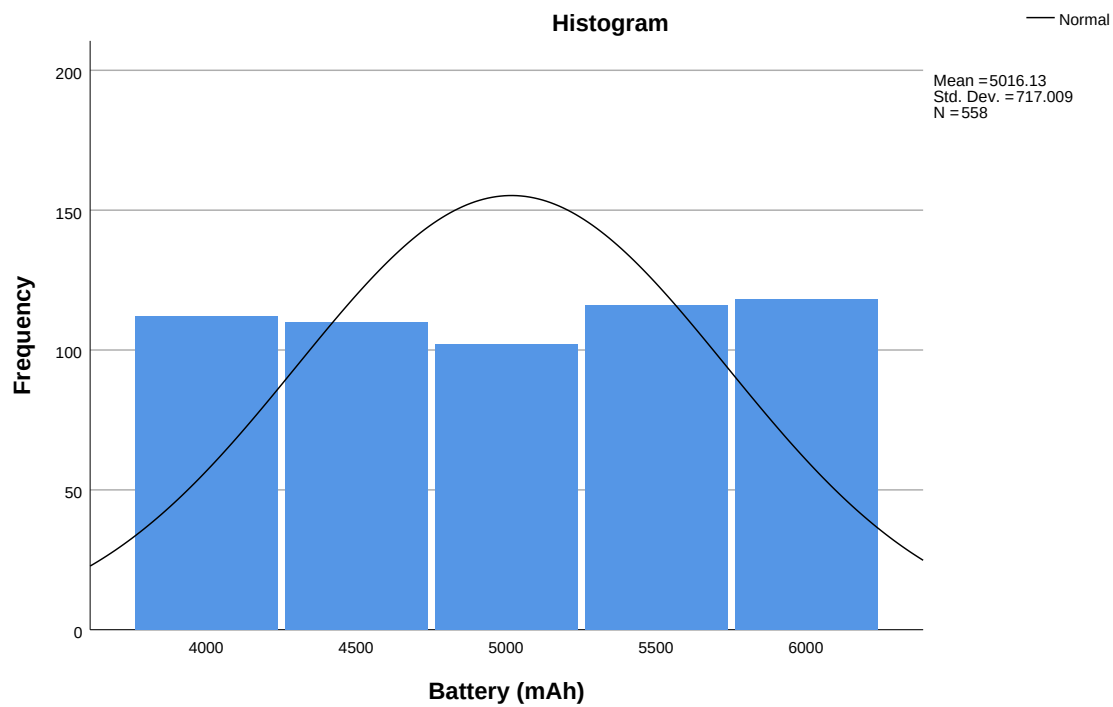


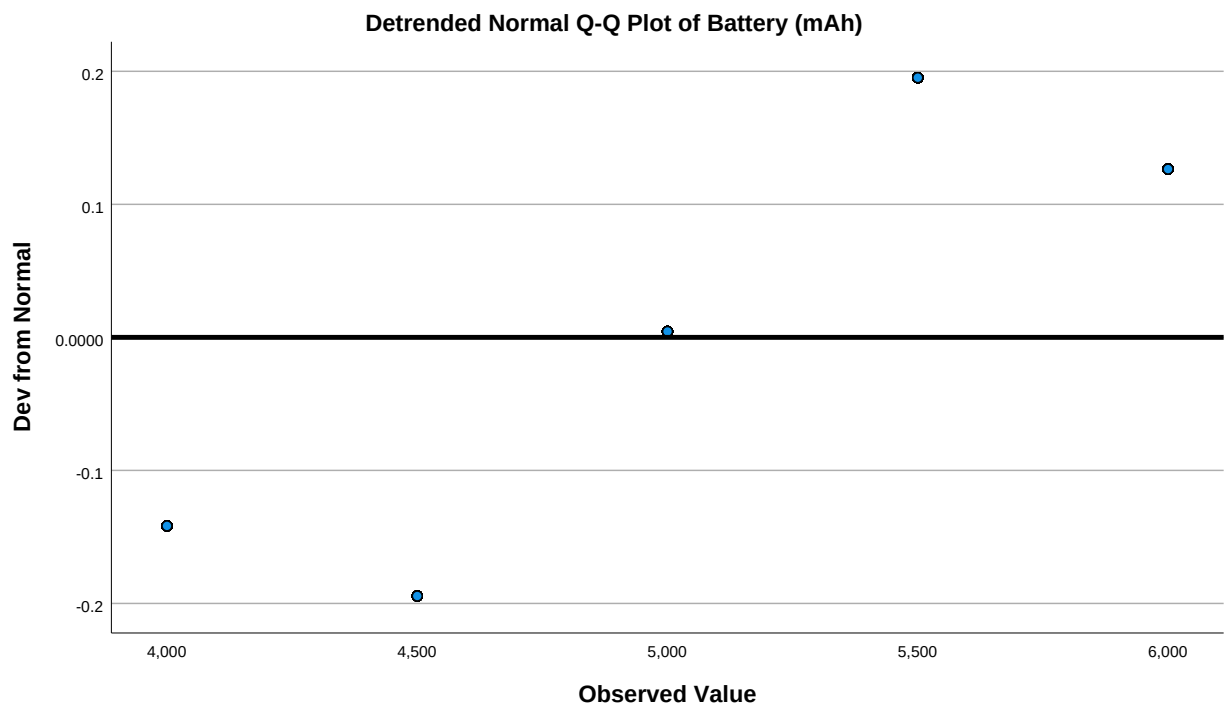
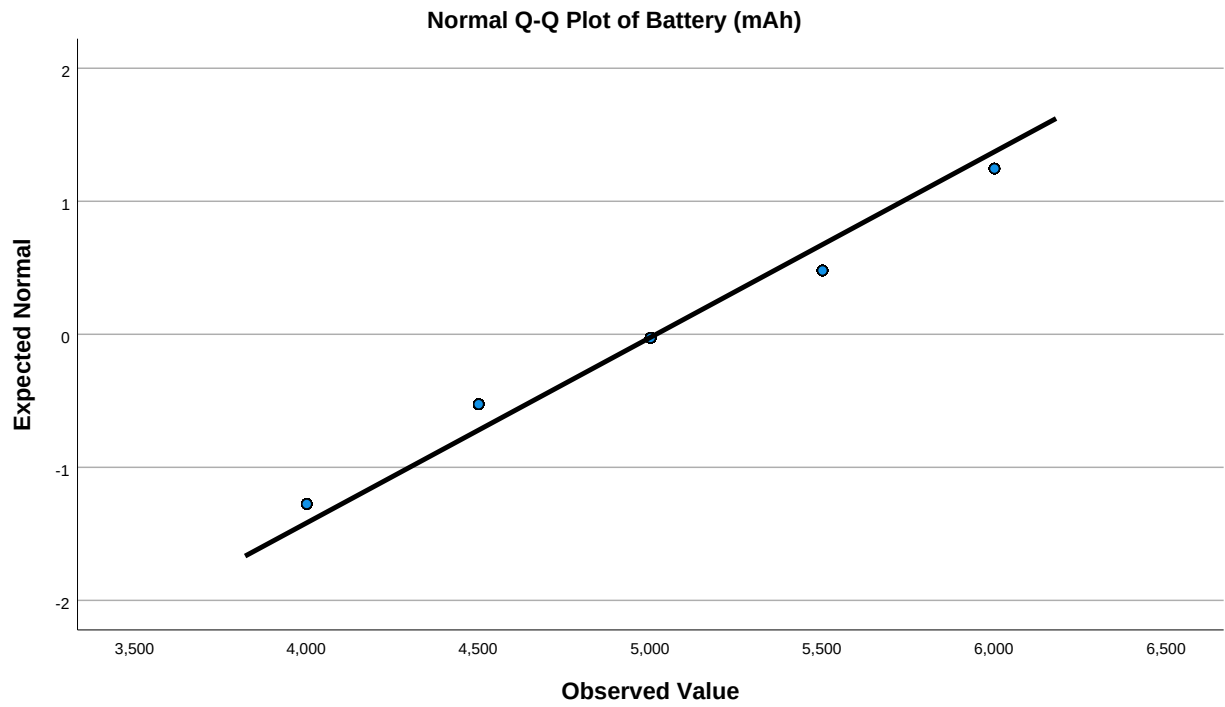
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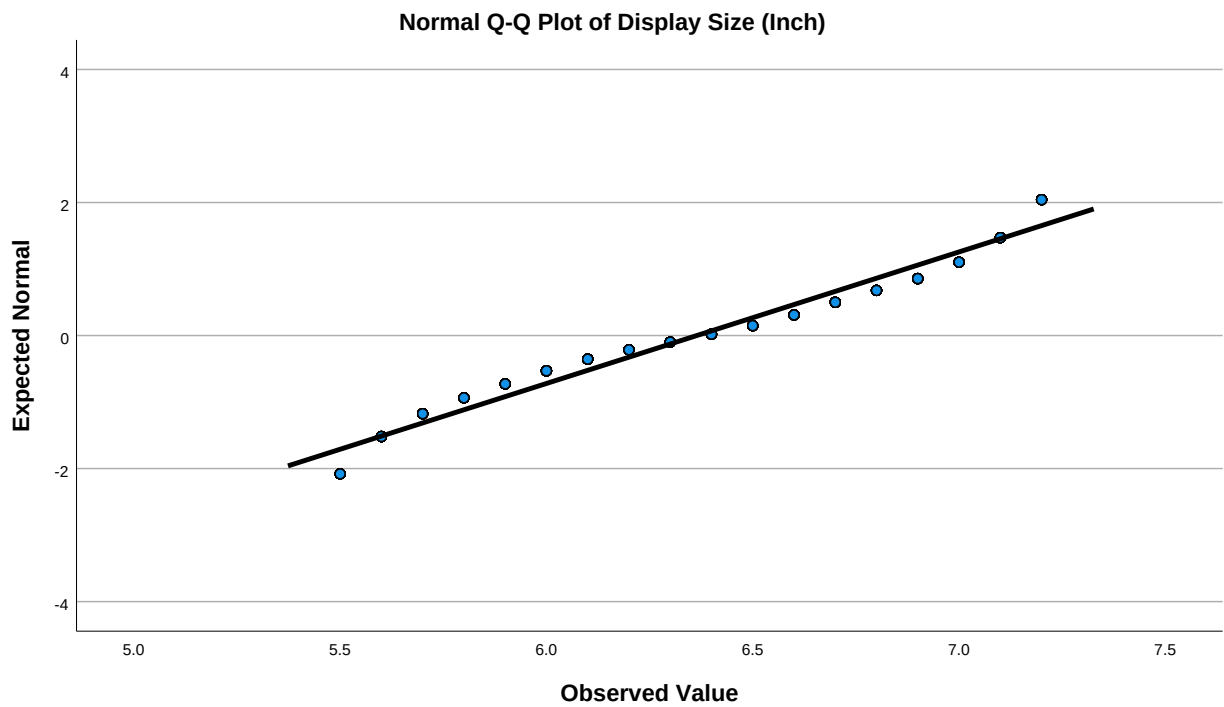
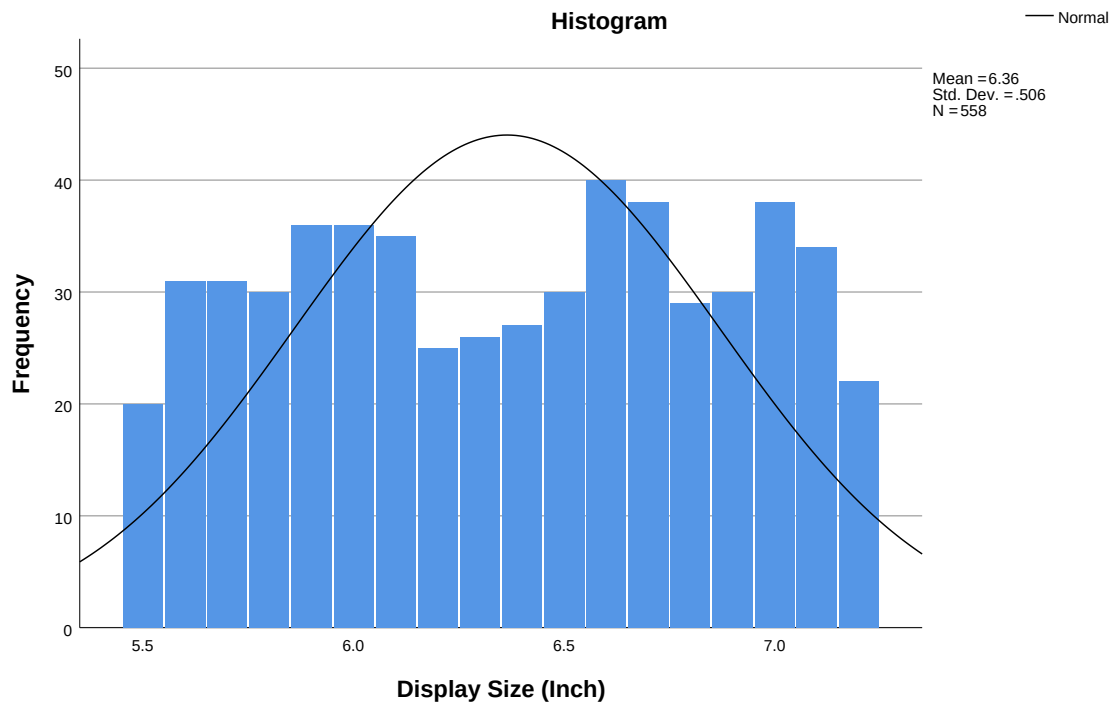


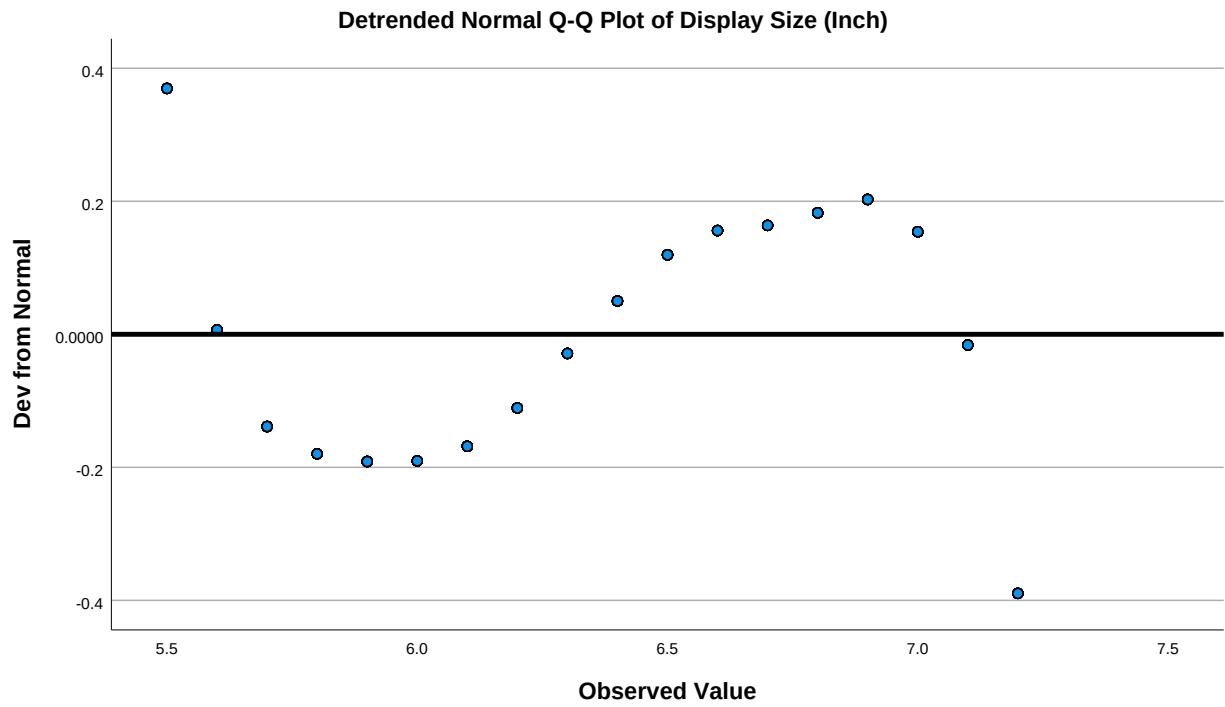
Battery (mAh)



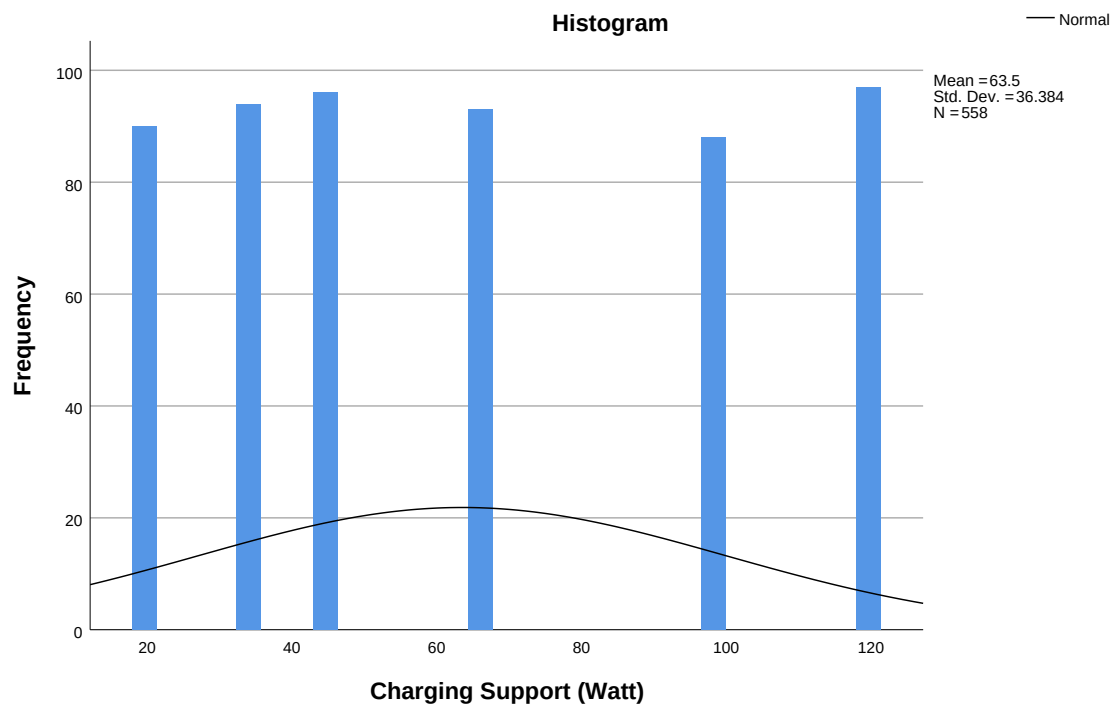


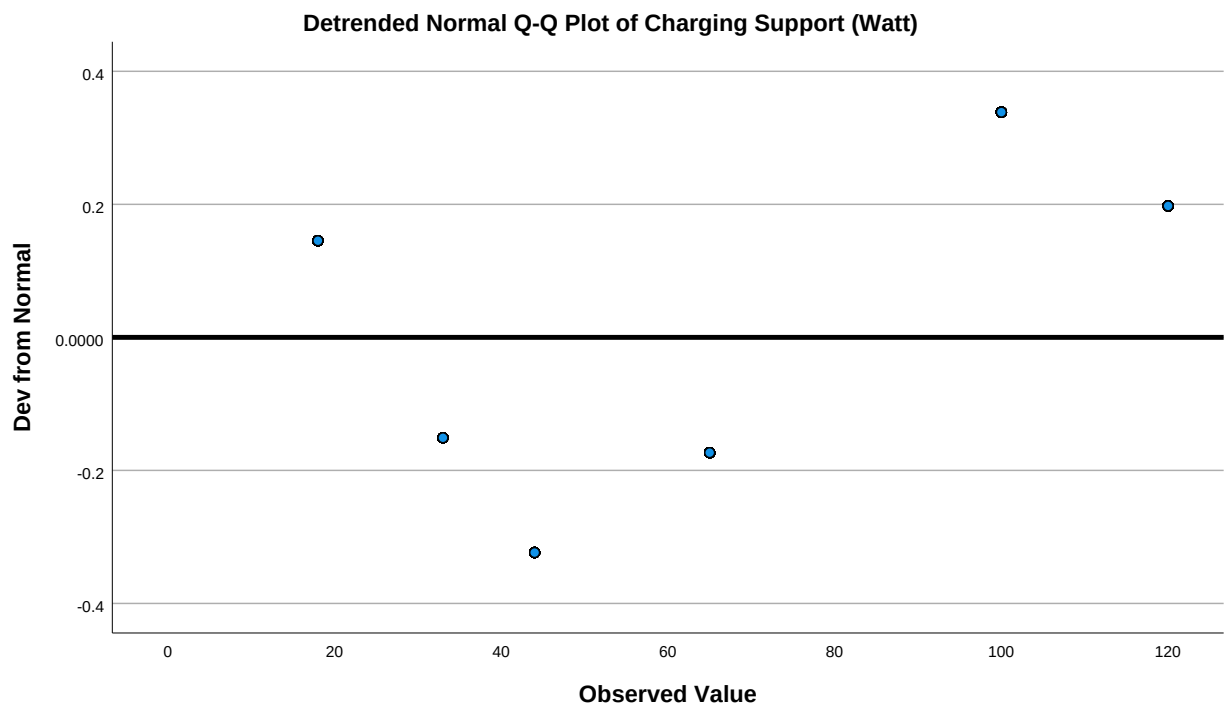
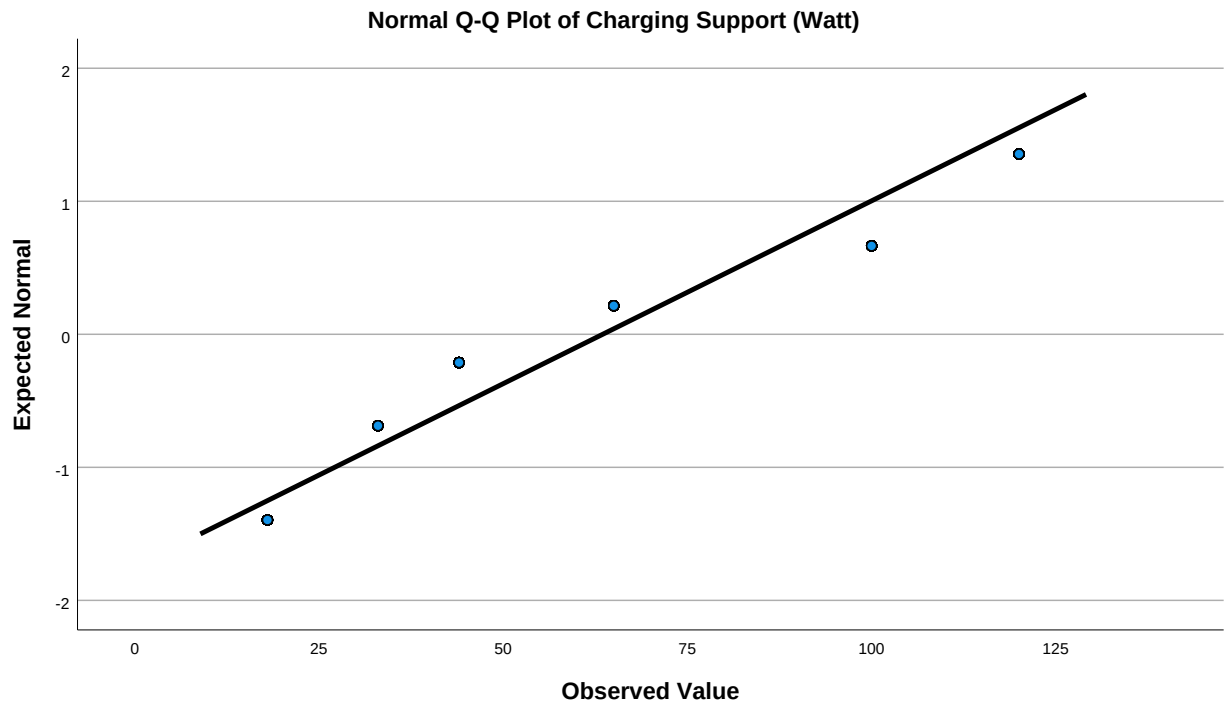
Display Size (Inch)





Charging Support (Watt)





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Frequencies

Notes

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	Cases Used	Statistics are based on all cases with valid data.
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Statistics

		Mobile Brand	5g Support	Operating system
N	Valid	558	558	558
	Missing	0	0	0

Frequency Table

Mobile Brand

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Apple	107	19.2	19.2	19.2
	Google	116	20.8	20.8	40.0
	OnePlus	116	20.8	20.8	60.8
	Samsung	105	18.8	18.8	79.6
	Xiaomi	114	20.4	20.4	100.0
	Total	558	100.0	100.0	

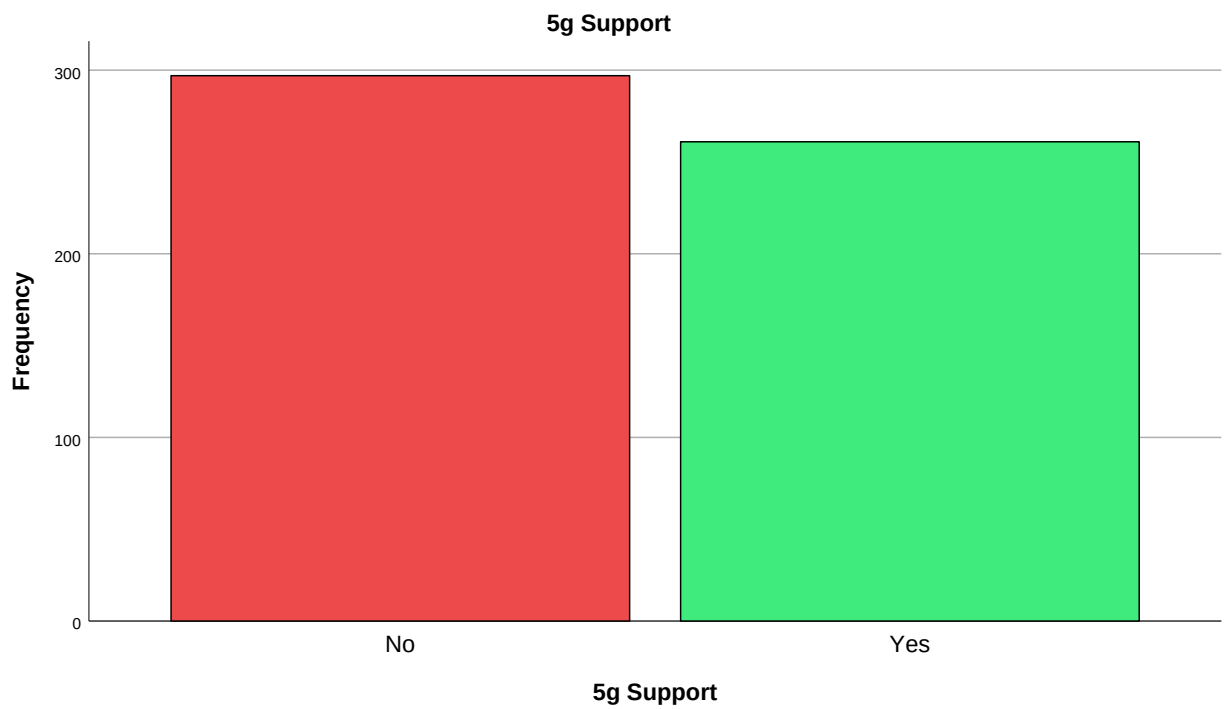
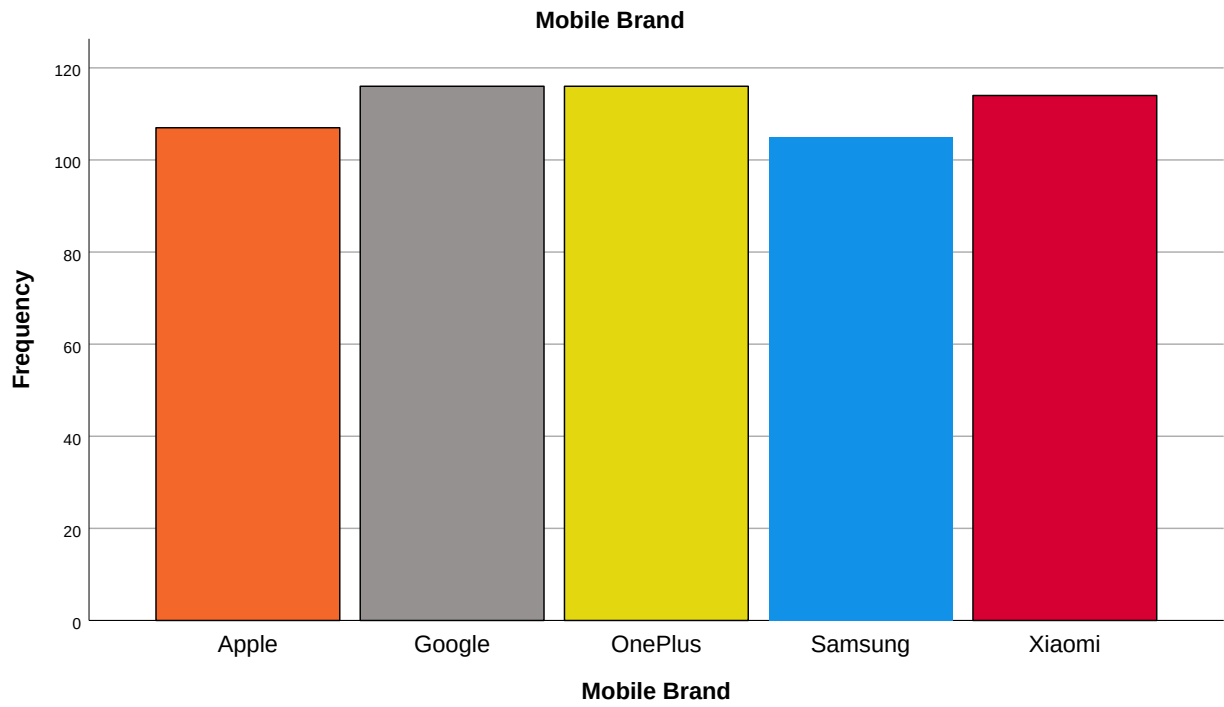
5g Support

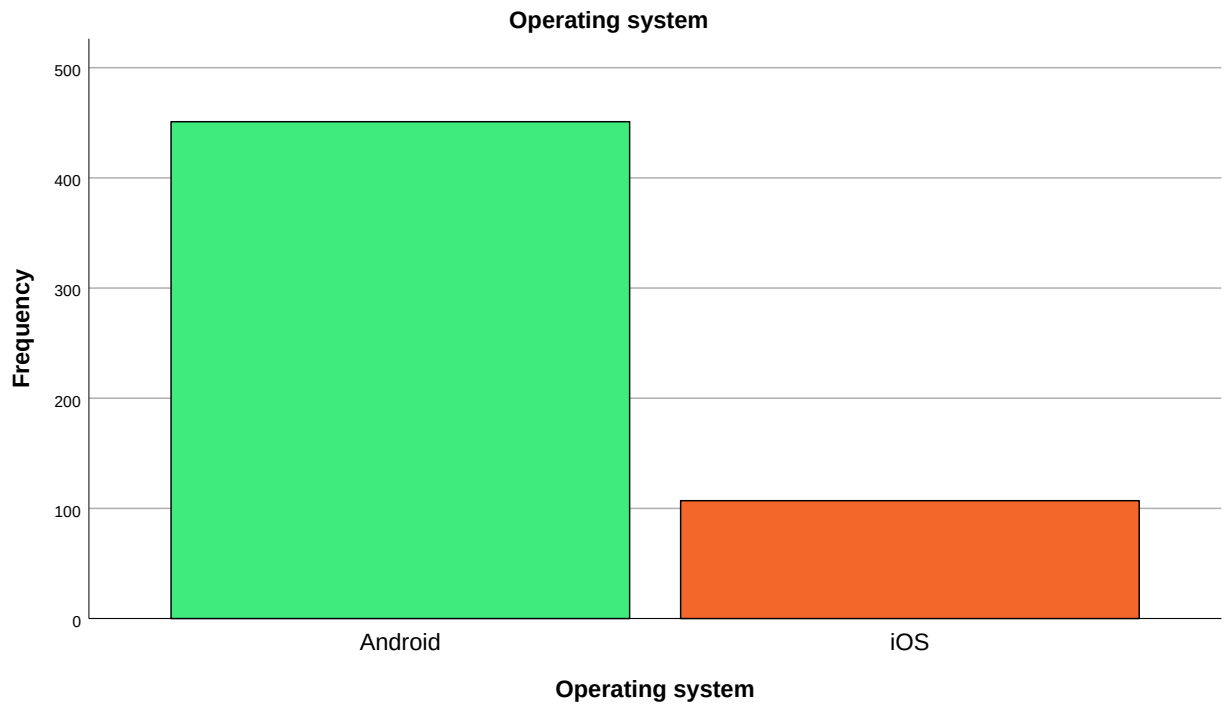
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	297	53.2	53.2	53.2
	Yes	261	46.8	46.8	100.0
	Total	558	100.0	100.0	

Operating system

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Android	451	80.8	80.8	80.8
	iOS	107	19.2	19.2	100.0
	Total	558	100.0	100.0	

Bar Chart





```
DESCRIPTIVES VARIABLES=price_usd ram_gb storage_gb camera_mp battery_mah display_size_inch
charging_watt rating
/STATISTICS=MEAN STDDEV VARIANCE RANGE MIN MAX KURTOSIS SKEWNESS.
```

Descriptives

Notes

Output Created		05-DEC-2025 01:41:36
Comments		
Input	Data	E:\BS Business Analytics\Fall 2025\Data Analysis 02\Semester Project\Final Folder\Project File.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	558
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	All non-missing data are used.
Syntax		DESCRIPTIVES VARIABLES=price_usd ram_gb storage_gb camera_mp battery_mah display_size_inch charging_watt rating /STATISTICS=MEAN STDDEV VARIANCE RANGE MIN MAX KURTOSIS SKEWNESS.
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.02

Descriptive Statistics

	N Statistic	Range Statistic	Minimum Statistic	Maximum Statistic	Mean Statistic	Std. Deviation Statistic
Price (\$)	558	1393	105	1498	815.18	410.963
RAM (GB)	558	12	4	16	9.08	4.253
ROM (GB)	558	960	64	1024	396.04	352.063
Camera (Megapixels)	558	188	12	200	82.57	60.782
Battery (mAh)	558	2000	4000	6000	5016.13	717.009
Display Size (Inch)	558	1.7	5.5	7.2	6.365	.5056
Charging Support (Watt)	558	102	18	120	63.50	36.384
Consumer Rating	558	1.5	3.5	5.0	4.220	.4409
Valid N (listwise)	558					

Descriptive Statistics

	Variance Statistic	Skewness		Kurtosis	
		Statistic	Std. Error	Statistic	Std. Error
Price (\$)	168890.536	-.068	.103	-1.222	.206
RAM (GB)	18.088	.405	.103	-1.193	.206
ROM (GB)	123948.063	.869	.103	-.709	.206
Camera (Megapixels)	3694.458	.997	.103	-.235	.206
Battery (mAh)	514102.044	-.035	.103	-1.340	.206
Display Size (Inch)	.256	-.031	.103	-1.229	.206
Charging Support (Watt)	1323.816	.371	.103	-1.333	.206
Consumer Rating	.194	.114	.103	-1.173	.206
Valid N (listwise)					

Notes

Output Created		05-DEC-2025 01:42:59
Comments		
Input	Data	E:\BS Business Analytics\Fall 2025\Data Analysis 02\Semester Project\Final Folder\Project File.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	558
Syntax		GRAPH /BAR(SIMPLE)=COUNT BY brand.
Resources	Processor Time	00:00:00.44
	Elapsed Time	00:00:00.31

```

ONEWAY price_usd BY Mobile_brand_code
/STATISTICS DESCRIPTIVES HOMOGENEITY
/PLOT MEANS
/MISSING ANALYSIS
/CRITERIA=CILEVEL(0.95)
/POSTHOC=TUKEY ALPHA(0.05).

```

Oneway

Notes

Output Created		05-DEC-2025 02:44:34
Comments		
Input	Data	E:\BS Business Analytics\Fall 2025\Data Analysis 02\Semester Project\Final Folder\Project File.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	558
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each analysis are based on cases with no missing data for any variable in the analysis.
Syntax		ONEWAY price_usd BY Mobile_brand_code /STATISTICS DESCRIPTIVES HOMOGENEITY /PLOT MEANS /MISSING ANALYSIS /CRITERIA=CILEVEL (0.95) /POSTHOC=TUKEY ALPHA(0.05).
Resources	Processor Time	00:00:00.59
	Elapsed Time	00:00:01.64

Descriptives

Price (\$)

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean	
					Lower Bound	Upper Bound
Apple	107	835.69	404.490	39.104	758.17	913.22
Google	116	808.06	403.923	37.503	733.77	882.35
OnePlus	116	812.25	431.788	40.090	732.84	891.66
Samsung	105	791.72	390.551	38.114	716.14	867.31
Xiaomi	114	827.74	426.293	39.926	748.64	906.84
Total	558	815.18	410.963	17.397	781.00	849.35

Descriptives

Price (\$)

	Minimum	Maximum
Apple	126	1477
Google	121	1480
OnePlus	118	1497
Samsung	142	1498
Xiaomi	105	1496
Total	105	1498

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
Price (\$)	Based on Mean	.679	4	553	.607
	Based on Median	.635	4	553	.638
	Based on Median and with adjusted df	.635	4	549.394	.638
	Based on trimmed mean	.675	4	553	.609

ANOVA

Price (\$)

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	127638.543	4	31909.636	.188	.945
Within Groups	93944390.25	553	169881.357		
Total	94072028.79	557			

Post Hoc Tests

Multiple Comparisons

Dependent Variable: Price (\$)

Tukey HSD

(I) Mobile Brand	(J) Mobile Brand	Mean Difference (I-J)	Std. Error	Sig.	95% ... Lower Bound
Apple	Google	27.631	55.246	.987	-123.57
	OnePlus	23.442	55.246	.993	-127.76
	Samsung	43.968	56.618	.937	-110.98
	Xiaomi	7.955	55.478	1.000	-143.88
Google	Apple	-27.631	55.246	.987	-178.83
	OnePlus	-4.190	54.120	1.000	-152.31
	Samsung	16.337	55.519	.998	-135.61
	Xiaomi	-19.676	54.357	.996	-168.44
OnePlus	Apple	-23.442	55.246	.993	-174.64
	Google	4.190	54.120	1.000	-143.93
	Samsung	20.526	55.519	.996	-131.42
	Xiaomi	-15.487	54.357	.999	-164.25
Samsung	Apple	-43.968	56.618	.937	-198.92
	Google	-16.337	55.519	.998	-168.28
	OnePlus	-20.526	55.519	.996	-172.47
	Xiaomi	-36.013	55.750	.967	-188.59
Xiaomi	Apple	-7.955	55.478	1.000	-159.79
	Google	19.676	54.357	.996	-129.09
	OnePlus	15.487	54.357	.999	-133.28
	Samsung	36.013	55.750	.967	-116.56

Multiple Comparisons

Dependent Variable: Price (\$)

Tukey HSD

95% Confidence .

(I) Mobile Brand	(J) Mobile Brand	Upper Bound
Apple	Google	178.83
	OnePlus	174.64
	Samsung	198.92
	Xiaomi	159.79
Google	Apple	123.57
	OnePlus	143.93
	Samsung	168.28
	Xiaomi	129.09
OnePlus	Apple	127.76
	Google	152.31
	Samsung	172.47
	Xiaomi	133.28
Samsung	Apple	110.98
	Google	135.61
	OnePlus	131.42
	Xiaomi	116.56
Xiaomi	Apple	143.88
	Google	168.44
	OnePlus	164.25
	Samsung	188.59

Homogeneous Subsets

Price (\$)

Tukey HSD^{a,b}

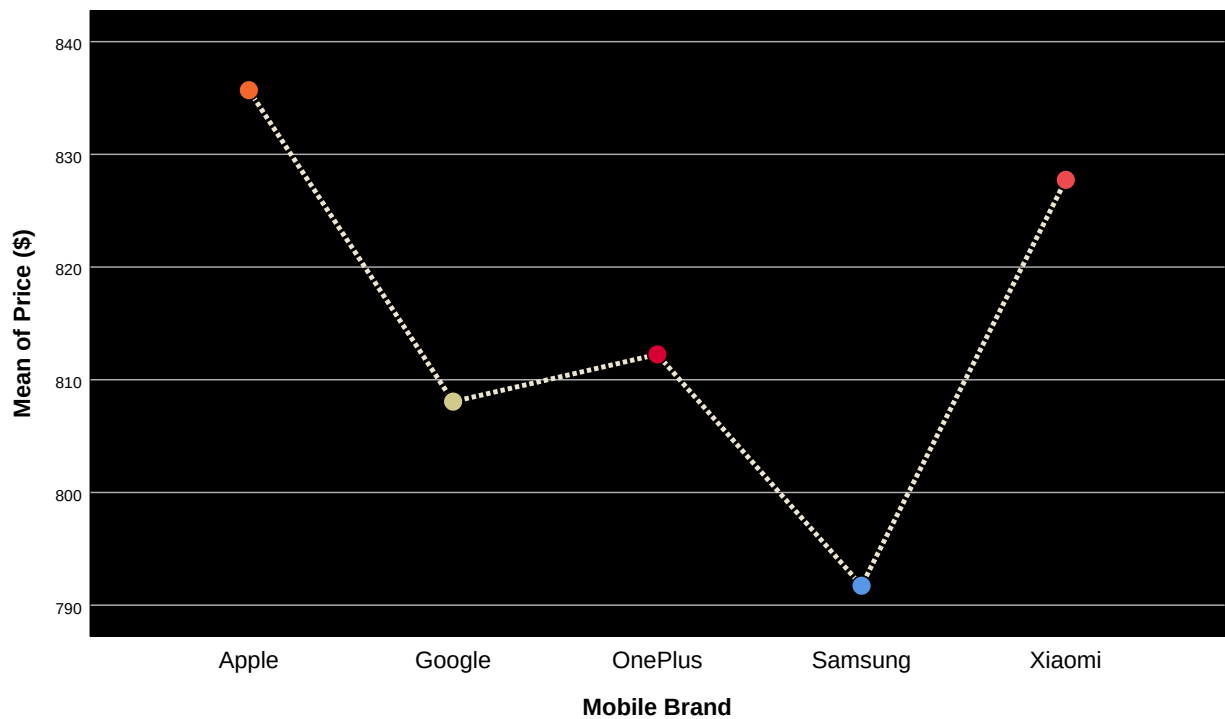
Mobile Brand	N	Subset for alpha = 0.05 1
Samsung	105	791.72
Google	116	808.06
OnePlus	116	812.25
Xiaomi	114	827.74
Apple	107	835.69
Sig.		.932

Means for groups in homogeneous subsets are displayed.

a. Uses Harmonic Mean Sample Size = 111.401.

b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

Means Plots



CORRELATIONS

```

/VARIABLES=price_usd ram_gb storage_gb camera_mp battery_mah display_size_inch charging
_watt
/PRINT=TWOTAIL NOSIG FULL

```

Correlations

Notes

Output Created		05-DEC-2025 02:56:35
Comments		
Input	Data	E:\BS Business Analytics\Fall 2025\Data Analysis 02\Semester Project\Final Folder\Project File.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	558
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each pair of variables are based on all the cases with valid data for that pair.
Syntax		CORRELATIONS /VARIABLES=price_usd ram_gb storage_gb camera_mp battery_mah display_size_inch charging_watt /PRINT=TWOTAIL NOSIG FULL /MISSING=PAIRWISE.
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.05

Correlations

		Price (\$)	RAM (GB)	ROM (GB)	Camera (Megapixels)
Price (\$)	Pearson Correlation	1	-.005	-.034	.030
	Sig. (2-tailed)		.906	.429	.475
	N	558	558	558	558
RAM (GB)	Pearson Correlation	-.005	1	.030	-.045
	Sig. (2-tailed)	.906		.477	.289
	N	558	558	558	558
ROM (GB)	Pearson Correlation	-.034	.030	1	.039
	Sig. (2-tailed)	.429	.477		.356
	N	558	558	558	558
Camera (Megapixels)	Pearson Correlation	.030	-.045	.039	1
	Sig. (2-tailed)	.475	.289	.356	
	N	558	558	558	558
Battery (mAh)	Pearson Correlation	-.010	.038	-.043	.032
	Sig. (2-tailed)	.812	.365	.315	.445
	N	558	558	558	558
Display Size (Inch)	Pearson Correlation	-.063	.022	-.008	-.112 ^{**}
	Sig. (2-tailed)	.139	.608	.853	.008
	N	558	558	558	558
Charging Support (Watt)	Pearson Correlation	.002	-.009	.027	.035
	Sig. (2-tailed)	.960	.832	.523	.403
	N	558	558	558	558

Correlations

		Battery (mAh)	Display Size (Inch)	Charging Support (Watt)
Price (\$)	Pearson Correlation	-.010	-.063	.002
	Sig. (2-tailed)	.812	.139	.960
	N	558	558	558
RAM (GB)	Pearson Correlation	.038	.022	-.009
	Sig. (2-tailed)	.365	.608	.832
	N	558	558	558
ROM (GB)	Pearson Correlation	-.043	-.008	.027
	Sig. (2-tailed)	.315	.853	.523
	N	558	558	558
Camera (Megapixels)	Pearson Correlation	.032	-.112**	.035
	Sig. (2-tailed)	.445	.008	.403
	N	558	558	558
Battery (mAh)	Pearson Correlation	1	.007	.009
	Sig. (2-tailed)		.878	.830
	N	558	558	558
Display Size (Inch)	Pearson Correlation	.007	1	.060
	Sig. (2-tailed)	.878		.156
	N	558	558	558
Charging Support (Watt)	Pearson Correlation	.009	.060	1
	Sig. (2-tailed)	.830	.156	
	N	558	558	558

** . Correlation is significant at the 0.01 level (2-tailed).

```

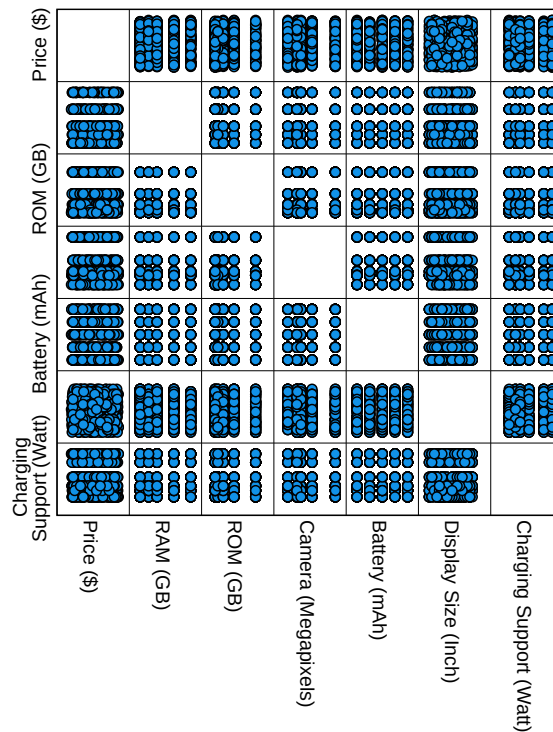
GRAPH
/SCATTERPLOT(MATRIX)=price_usd ram_gb storage_gb camera_mp battery_mah display_size_inch
charging_watt
/MISSING=LISTWISE.

```

Graph

Notes

Output Created		05-DEC-2025 03:03:47
Comments		
Input	Data	E:\BS Business Analytics\Fall 2025\Data Analysis 02\Semester Project\Final Folder\Project File.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	558
Syntax		GRAPH /SCATTERPLOT (MATRIX)=price_usd ram_gb storage_gb camera_mp battery_mah display_size_inch charging_watt /MISSING=LISTWISE.
Resources	Processor Time	00:00:00.48
	Elapsed Time	00:00:00.69



```

T-TEST GROUPS=support_5g(1 2)
/MISSING=ANALYSIS
/VARIABLES=price_usd
/ES DISPLAY(TRUE)
/CRITERIA=CI(.95).

```

T-Test

Notes

Output Created		05-DEC-2025 03:11:47
Comments		
Input	Data	E:\BS Business Analytics\Fall 2025\Data Analysis 02\Semester Project\Final Folder\Project File.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	558
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	Statistics for each analysis are based on the cases with no missing or out-of-range data for any variable in the analysis.
Syntax		T-TEST GROUPS=support_5g(1 2) /MISSING=ANALYSIS /VARIABLES=price_usd /ES DISPLAY(TRUE) /CRITERIA=CI(.95).
Resources	Processor Time	00:00:00.03
	Elapsed Time	00:00:00.05

Group Statistics

	5g Support	N	Mean	Std. Deviation	Std. Error Mean
Price (\$)	No	297	833.04	402.052	23.329
	Yes	261	794.85	420.724	26.042

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means	
		F	Sig.	t	df
Price (\$)	Equal variances assumed	.878	.349	1.095	556
	Equal variances not assumed			1.092	539.538

Independent Samples Test

		t-test for Equality of Means			
		Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference Lower Upper
Price (\$)	Equal variances assumed	.274	38.186	34.861	-30.289
	Equal variances not assumed	.275	38.186	34.964	-30.495

Independent Samples Test

		t-test for Equality of Means
		95% Confidence Interval of the Difference Upper
Price (\$)	Equal variances assumed	106.662
	Equal variances not assumed	106.868

Independent Samples Effect Sizes

				95% Confidence Interval	
Standardizer ^a			Point Estimate	Lower	Upper
Price (\$)	Cohen's d	410.889	.093	-.073	.259
	Hedges' correction	411.445	.093	-.073	.259
	Glass's delta	420.724	.091	-.076	.257

a. The denominator used in estimating the effect sizes.

Cohen's d uses the pooled standard deviation.

Hedges' correction uses the pooled standard deviation, plus a correction factor.

Glass's delta uses the sample standard deviation of the control group.

T - TEST

/TESTVAL=4

/MISSING=ANALYSIS

/VARIABLES=rating

/ES DISPLAY(TRUE)

/CRITERIA=CI(.95).

T-Test

Notes

Output Created		05-DEC-2025 03:15:59
Comments		
Input	Data	E:\BS Business Analytics\Fall 2025\Data Analysis 02\Semester Project\Final Folder\Project File.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	558
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	Statistics for each analysis are based on the cases with no missing or out-of-range data for any variable in the analysis.

Notes

Syntax	T-TEST /TESTVAL=4 /MISSING=ANALYSIS /VARIABLES=rating /ES DISPLAY(TRUE) /CRITERIA=CI(.95).	
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
Consumer Rating	558	4.220	.4409	.0187

One-Sample Test

Test Value = 4					
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference Lower Upper
Consumer Rating	11.809	557	.000	.2204	.184

One-Sample Test

Test Value = 4	
95% Confidence Interval of the Difference	
	Upper Lower
Consumer Rating	.257

One-Sample Effect Sizes

		Standardizer ^a	Point Estimate	95% Confidence Interval	
				Lower	Upper
Consumer Rating	Cohen's d	.4409	.500	.412	.588
	Hedges' correction	.4415	.499	.411	.587

- a. The denominator used in estimating the effect sizes.
Cohen's d uses the sample standard deviation.
Hedges' correction uses the sample standard deviation, plus a correction factor.

```

/MISSING LISTWISE
/STATISTICS COEFF OUTS R ANOVA COLLIN TOL
/CRITERIA=PIN(.05) POUT(.10)
/NOORIGIN
/DEPENDENT price_usd
/METHOD=ENTER ram_gb storage_gb camera_mp battery_mah display_size_inch charging_watt
/SCATTERPLOT=( *ZRESID , *ZPRED) .

```

Regression

Notes

Output Created		05-DEC-2025 03:28:10
Comments		
Input	Data	E:\BS Business Analytics\Fall 2025\Data Analysis 02\Semester Project\Final Folder\Project File.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	558
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA COLLIN TOL /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT price_usd /METHOD=ENTER ram_gb storage_gb camera_mp battery_mah display_size_inch charging_watt /SCATTERPLOT= (*ZRESID ,*ZPRED).

Notes

Resources	Processor Time	00:00:00.45
	Elapsed Time	00:00:00.47
	Memory Required	6080 bytes
	Additional Memory Required for Residual Plots	144 bytes

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	Charging Support (Watt), RAM (GB), Battery (mAh), ROM (GB), Display Size (Inch), Camera (Megapixels) ^b	.	Enter

a. Dependent Variable: Price (\$)

b. All requested variables entered.

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.077 ^a	.006	-.005	411.978

a. Predictors: (Constant), Charging Support (Watt), RAM (GB), Battery (mAh), ROM (GB), Display Size (Inch), Camera (Megapixels)

b. Dependent Variable: Price (\$)

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	553084.453	6	92180.742	.543	.775 ^b
	Residual	93518944.34	551	169725.852		
	Total	94072028.79	557			

a. Dependent Variable: Price (\$)

b. Predictors: (Constant), Charging Support (Watt), RAM (GB), Battery (mAh), ROM (GB), Display Size (Inch), Camera (Megapixels)

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1161.443	258.454		4.494	.000
	RAM (GB)	-.095	4.115	-.001	-.023	.982
	ROM (GB)	-.042	.050	-.036	-.837	.403
	Camera (Megapixels)	.169	.290	.025	.584	.559
	Battery (mAh)	-.007	.024	-.012	-.283	.777
	Display Size (Inch)	-49.111	34.824	-.060	-1.410	.159
	Charging Support (Watt)	.067	.481	.006	.139	.889

Coefficients^a

Model		Collinearity Statistics	
		Tolerance	VIF
1	(Constant)		
	RAM (GB)	.995	1.005
	ROM (GB)	.995	1.005
	Camera (Megapixels)	.981	1.019
	Battery (mAh)	.995	1.005
	Display Size (Inch)	.983	1.017
	Charging Support (Watt)	.994	1.006

a. Dependent Variable: Price (\$)

Collinearity Diagnostics^a

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions		
				(Constant)	RAM (GB)	ROM (GB)
1	1	5.911	1.000	.00	.00	.01
	2	.406	3.818	.00	.00	.93
	3	.322	4.287	.00	.05	.03
	4	.224	5.141	.00	.22	.01
	5	.121	6.995	.00	.71	.01
	6	.014	20.252	.03	.00	.01
	7	.003	46.090	.97	.00	.00

Collinearity Diagnostics^a

Model	Dimension	Variance Proportions			
		Camera (Megapixels)	Battery (mAh)	Display Size (Inch)	Charging Support (Watt)
1	1	.01	.00	.00	.01
	2	.06	.00	.00	.02
	3	.83	.00	.00	.06
	4	.00	.00	.00	.74
	5	.07	.03	.01	.17
	6	.00	.87	.12	.01
	7	.02	.10	.88	.00

a. Dependent Variable: Price (\$)

Residuals Statistics^a

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	740.12	897.07	815.18	31.511	558
Residual	-741.222	745.909	.000	409.753	558
Std. Predicted Value	-2.382	2.599	.000	1.000	558
Std. Residual	-1.799	1.811	.000	.995	558

a. Dependent Variable: Price (\$)

Charts

